

PikkuFold: Efficient Folding in a Few Kilobytes

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Abstract. Folding is a powerful technique for constructing efficient succinct proof systems, especially for computations expressed in a streaming fashion. We present **PikkuFold**, a new lattice-based folding protocol that improves upon state-of-the-art folding schemes such as SALSAA (ePrint 2025/2124) and Cyclo (EUROCRYPT 2026). One folding step communicates 5.7 KB in addition to the commitments to its fresh inputs, compared with ≥ 30 KB for Cyclo and ≥ 60 KB for SALSAA on similar instances, with a prover more than an order of magnitude faster and the verifier in the millisecond range. At the heart of our construction are *layered random projections*, whose algebraic structure makes them fast to verify and whose final image is short enough to send to the verifier directly, eliminating the cost of auxiliary commitments. We use these techniques to replace Cyclo’s costly and restrictive range proofs while keeping the increase in the accumulator norm small and additive across multiple folds. **PikkuFold** is the first lattice-based construction that does not require any in-protocol commitments beyond those of the fresh input relations. Such commitments account for much of a folding transcript: every prior lattice-based scheme commits to a decomposed or otherwise transformed witness during the fold, immediately increasing the communication by dozens of kilobytes. We also provide two contributions of independent interest beyond folding schemes:

- (i) a Johnson–Lindenstrauss theorem for biased ternary matrices modulo q with certified concrete constants, which replaces the heuristic parametrisation used in prior works, and
- (ii) a thorough analysis of the short-challenge sampler with fixed Hamming weight and operator-norm rejection, offering a wide range of parameter sets. Using this sampler as a drop-in replacement would lead to immediate improvements across a wide range of lattice-based protocols.

1 Introduction

Succinct non-interactive arguments of knowledge (SNARKs) have emerged as a key primitive in modern cryptography, finding applications in blockchains [XZC⁺22, GKO24, LM25, LZW⁺25], privacy-preserving authentication [OREHS24], and verifiable computation [ACGS24, CCC⁺25]. However, a fundamental memory bottleneck limits their practical deployment [LZW⁺24, ZSCZ25]: generating a proof requires the prover to hold the entire computation in memory simultaneously. For large computations, this is often infeasible, and it fails to exploit the fact that many real-world computations proceed in a streaming or incremental fashion.

Practical scenarios such as blockchain state transitions and verifiable outsourced computation (e.g., cloud/edge ML inference and long-running services) call for a more scalable approach. This has motivated the study of *incrementally verifiable computation* (IVC) [Val08, NPR19] and the more general notion of proof-carrying data [CT10, BCL⁺21, BDFG21, CCG⁺23]. Early IVC constructions relied on SNARKs, recursively producing

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at each step a proof that the previous verifier accepted [Val08, BCCT12, BCTV14]. A more recent and efficient paradigm is that of *folding schemes* [BGH19, BDFG21, BCL⁺21, BC25a, BC25b], where the prover compresses many computation steps into a single accumulator instance, avoiding the overhead of full recursive proof verification at every step.

Recent constructions tend to present folding schemes as a form of reduction of knowledge (RoK) [ACK21, KP22]. A reduction of knowledge is an interactive protocol between a prover and a verifier that reduces an *instance* of an input relation Ξ^{input} to an *instance* of an output relation Ξ^{output} . For soundness, or more precisely knowledge soundness, we consider an extractor that, given a prover that can convince the verifier to accept with non-negligible probability, can extract a witness for an input relation $\Xi^{\text{input}'}$ from a witness for an output relation $\Xi^{\text{output}'}$. Notably, $\Xi^{\text{input}'}$ may differ from Ξ^{input} , and $\Xi^{\text{output}'}$ may differ from Ξ^{output} . In practice, this means that the extractor may recover a *relaxed* witness for a *relaxed* relation, and this relaxation may propagate through a chain of RoKs.

In this language, a folding scheme is a form of RoK in which the input relation is $\Xi^{\text{input}} := (\Xi^{\text{acc}}, (\Xi_i^{\text{input}})_{i \in [L]})$ and the output relation is $\Xi^{\text{output}} := \Xi^{\text{acc}'}$. Here, Ξ^{acc} is an accumulator relation, and $(\Xi_i^{\text{input}})_{i \in [L]}$ is a sequence of L input relations, e.g. relations certifying the correctness of a computation. The folding scheme takes as input a witness for the accumulator relation Ξ^{acc} and witnesses for the input relations $(\Xi_i^{\text{input}})_{i \in [L]}$, and outputs a witness for the related accumulator relation $\Xi^{\text{acc}'}$.

The best-established line of work uses additively homomorphic group commitments, typically Pedersen commitments, and includes Nova, SuperNova, HyperNova, Protostar, ProtoGalaxy, Mangrove, Mova, and MicroNova [KST22, KS22, KS24, BC23, EG23, NDC⁺24, DGMV24, ZSCZ25]. These constructions ultimately rely on discrete-logarithm-type assumptions and are therefore not post-quantum secure. Another line of work constructs hash-based accumulation schemes, closely related to folding schemes, from Merkle commitments and error-correcting codes [BMNW25a, BMNW25b, BCFW25]. Although they avoid “public-key assumptions”, these constructions require codeword encodings and numerous Merkle authentication paths, leading to relatively large communication. Their concrete efficiency also remains less thoroughly explored.

1.1 Lattice-Based Folding Schemes

A major recent advance is the development of lattice-based folding schemes, including Lova [FKNP24], LatticeFold [BC25a], LatticeFold+ [BC25b], Neo and SuperNeo [NS26], SALSAA [KLOT25], Cyclo [GLLO26], and ProtogaLattice [BNP26].

The special role of lattice-based cryptography in the construction of modern folding schemes stems from the availability of simple linearly homomorphic commitments. At the heart of each of the aforementioned schemes is an Ajtai-style commitment scheme [Ajt96], whose security is based on the Short Integer Solution (SIS) problem. A commitment to a short vector $\mathbf{w} \in \mathcal{R}_q^m$ is a vector $\mathbf{y} \in \mathcal{R}_q^n$ such that

$$\begin{aligned} \mathbf{A}\mathbf{w} &= \mathbf{y} \pmod{q} \\ \|\mathbf{w}\| &\leq \beta, \end{aligned} \tag{1}$$

where $\mathcal{R} \cong \mathbb{Z}[x]/\langle \Phi_f \rangle$ is a cyclotomic ring, $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ for some (typically prime) modulus q , and $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ is a public random matrix.

Each of the above lattice-based folding schemes uses a similar approach to folding, which we summarise as follows. The prover holds a sequence of L commitments $\mathbf{y}_0, \dots, \mathbf{y}_{L-1}$ to short vectors $\mathbf{w}_0, \dots, \mathbf{w}_{L-1}$, together with an accumulator commitment $\mathbf{y}_{\text{acc}}$ to a short vector $\mathbf{w}_{\text{acc}}$. The parties then use the verifier’s short challenges to compute a linear combination of the commitments $\mathbf{y}_i$ and the accumulator commitment $\mathbf{y}_{\text{acc}}$, obtaining a new accumulator commitment $\mathbf{y}'_{\text{acc}}$. Likewise, the prover computes a linear combination of

the short vectors $\mathbf{w}_i$ and the accumulator witness $\mathbf{w}_{\text{acc}}$, obtaining a new witness $\mathbf{w}'_{\text{acc}}$. This extremely simple approach comes at the price of a gradual *degradation* of the accumulator norm in both the *correctness* and *extraction* directions. Most work in this area addresses these two problems. Prior to Cyclo [GLLO26], the proposed solutions can be summarised as two families of subroutines.

Norm growth in the *correctness* direction is typically addressed via *decomposition*. This approach, exemplified in [FKNP24, BC25a, BC25b, NS26, KLOT25] and, in part, in [BNP26], requires the prover to decompose the folded witness $\mathbf{w}'_{\text{acc}}$ according to some decomposition basis and then recommit to the decomposed witness. This simple approach comes at the cost of increased prover communication.

On the other hand, prior work offers at least four approaches to norm growth in the *extraction* direction:

- (A1) *Range proof via polynomial identity.* This folklore approach, made concrete in [BC25a, NS26, GLLO26], proves that each coefficient of the folded witness $\mathbf{w}'_{\text{acc}}$ is bounded by some β , or equivalently that its *coefficient ℓ_∞ -norm* is at most β . This claim can be expressed as a polynomial identity and proved using the sumcheck protocol. However, all three costs, i.e. communication, prover time, and verifier time, are linear in β , making this approach impractical for superconstant β .
- (A2) *Range proof via monomial decomposition.* Another interesting approach, described in [BC25b], decomposes the folded witness $\mathbf{w}'_{\text{acc}}$ into a linear combination of monomials (i.e. elements of the power basis of the cyclotomic ring $\mathcal{R}$) and then proves the correctness of the decomposition. This approach exploits the bound imposed by the *degree* of $\mathcal{R}$. The price is that the prover must send a commitment to the decomposition and invoke an additional round of sumcheck. Moreover, as shown in [GLLO26], computing commitments to the decomposition is computationally expensive.
- (A3) *Approximate proof via Johnson–Lindenstrauss.* Yet another approach, informally described in [Osa26] and treated more formally in [KLOT25], uses Johnson–Lindenstrauss projections, which are norm-preserving in the sense that $\|\mathbf{J}\mathbf{w}\| \approx \|\mathbf{w}\|$, in a structured variant that admits a succinct description. Although $\mathbf{J}\mathbf{w}$ has a smaller dimension than $\mathbf{w}$, it is still too long to send to the verifier. The prover therefore commits to $\mathbf{J}\mathbf{w}$ and proves that the commitment is correct. The resulting relation cannot be folded immediately, so the accumulator must contain at least two commitments, even before accounting for decomposition.
- (A4) *Approximate norm-preserving extraction via subtractive set.* *Subtractive sets* [AL21] have not been used for folding schemes, but they can serve as the extractor’s challenge set, yielding a witness $\mathbf{w}'_{\text{acc}}$ such that $\|\mathbf{w}'_{\text{acc}}\| \approx \|\mathbf{w}_{\text{acc}}\|$. This approach was used in lattice-based SNARKs [AL21, CLM23, KLNO24]. However, such sets are very small, necessitating amplification of the folding, which again increases the size of the accumulator relation.

Approaches (A3) and (A4) incur *extraction norm growth*: the extracted witness is slightly larger, by a multiplicative factor, than the original witness. Norm-upgrading techniques [LNP22, KLNO24, KLOT25] address this problem by proving the exact norm of a witness already known to satisfy a relaxed bound $\beta' \geq \beta$.

Cyclo [GLLO26] makes a complementary observation. It shows that, instead of choosing between multiplicative norm growth in the extraction direction – which becomes exponential in the number of folding steps – and an expensive range proof for the accumulator, one can prove a norm bound only for the input relations (not the accumulator) and accept additive norm growth, which is only linear in the number of folding steps. Consequently, norm growth in the correctness direction also becomes additive, eliminating the need for

decomposition for a bounded number of folding steps. The *trick of Cyclo* is simply to avoid combining the accumulator commitment with any challenge. This requires the accumulator to remain a *single* relation of a form similar to Equation (1).

Therefore, to take advantage of these optimisations, Cyclo applies approach (A1) to fresh inputs. This limits its applicability to relations whose coefficient ℓ_∞ -norm bound β is sufficiently small. For larger bounds, Cyclo resorts to a relatively expensive *extension commitment*, which is essentially a recommitment to an aggressively decomposed witness. At the same time, the single-relation accumulator required by Cyclo cannot directly carry the additional projection relation introduced by approach (A3). Thus, Cyclo provides additive norm growth without efficient support for large-norm inputs, whereas projection-based approaches support large norms at the cost of an additional commitment and accumulator relation.

This raises the following question:

Is it possible to construct a lattice-based folding scheme that simultaneously achieves:

- (i) *norm preservation*, up to small additive growth in both the correctness and extraction directions,
- (ii) *communication efficiency*, in the sense that the prover sends no additional Ajtai commitments during folding, and
- (iii) *range-check-free efficiency*, in the sense that the cost of folding does not depend linearly on the coefficient ℓ_∞ -norm bound β ?

1.2 Our Contributions

In this work, we answer the above question in the affirmative. We present **PikkuFold**, a new lattice-based folding scheme that achieves all three properties simultaneously. Our contributions are as follows.

- (C1) We construct a folding scheme based on *layered random projections*. Each layer is structured enough for efficient verification and reduces the witness dimension. After the final layer, the projection image is short enough to send directly to the verifier. The layer-correctness claims admit a multilinear description and are verified by a single ring-sumcheck invocation. Consequently, the prover sends no additional commitment to the projection image, and the accumulator remains a single relation compatible with additive norm growth.
- (C2) We prove a Johnson–Lindenstrauss theorem for biased ternary matrices under modular reduction. Previous concrete parameters for such matrices were justified heuristically [GHL22, BS23]. Our theorem supplies the rigorous tail bounds needed to instantiate the projections of (C1) without this heuristic, and it does so at a cost of only a few percent in the resulting norm-preservation constants.
- (C3) We construct and analyse an efficient sampler for low-norm, fixed-weight ternary challenges. The sampler first draws from the exact fixed-weight distribution and then rejects elements whose multiplication operator norm is too large. Its rejection threshold exposes an explicit tradeoff between challenge norm and sampling time. We additionally give a heuristic variant for almost-splitting cyclotomic rings.
- (C4) We provide concrete parameter selections for **PikkuFold** and a vectorised implementation at specialised parameters. Concretely, the folding protocol communicates roughly 5.7 KB, excluding the fresh input commitments, while retaining millisecond verification. At matched input sizes, the prover is more than an order of magnitude

faster than the SALSAA implementation on the same machine. Thus, once the inputs are committed, an entire lattice-based fold fits in only a few kilobytes of communication.

The contributions above are modular. The folding scheme of (C1) is analysed through abstract guarantees on the projection and challenge distributions. Contributions (C2) and (C3) establish suitable guarantees using largely independent techniques and may also be used outside folding schemes, for example in lattice-based SNARKs. They are combined only when selecting concrete parameters and evaluating (C4).

Although we state our folding scheme for polynomial-evaluation relations, it serves as a building block for more expressive lattice-based constructions. In particular, it yields a folding scheme for AIR relations (Section 7.2) with little additional machinery. Moreover, the bounded folding depth caused by additive norm growth can be extended by periodically performing decomposition and an exact norm check, a costly operation amortised over the folding steps (Section 7.1).

1.3 Perspective

At our concrete parameters, the folding messages excluding fresh input commitments occupy a few kilobytes, and the Ajtai commitments now dominate total communication. This shifts the main bottleneck from the folding protocol to the commitment layer. A natural next step is therefore to combine `PikkuFold` with techniques for smaller, recursively composed commitments, in the spirit of [BS23, BBL⁺26a, KLN⁺26].

A second open direction is zero knowledge: the presented protocol is not zero knowledge, and hiding the witness throughout sumcheck without losing its communication and prover efficiency requires additional techniques, for example those developed in Libra [XZZ⁺19].

We envision using `PikkuFold` as a building block for incrementally verifiable computation (IVC) [Val08, KST22]. Informally, an IVC folds not only the witness but also the verifier’s execution of the preceding fold, covering both its algebraic claims and the correctness of the hash execution, assuming a Fiat–Shamir transform. Such an arithmetisation of the hash function is a nontrivial task. The arithmetisation of *standard* hashes such as SHA-256, BLAKE3 and SHA-3 has recently become an active research area [DP25, Bin26, BRW26]. These works use polynomial commitments over binary fields or field-agnostic polynomial commitments, such as those of [ZCF24, Dia25, QGZ26, DP26]. Porting these techniques to the lattice setting is an interesting open problem.

1.4 Paper Organisation

The three main technical components use largely independent techniques, so a single technical overview would obscure their modularity. We therefore present each component in its own self-contained section. In Section 3, we construct and analyse the folding scheme under abstract guarantees on the projection and challenge distributions. In Section 4, we prove the modular Johnson–Lindenstrauss theorem used to instantiate the projections, while Section 5 develops the low-norm challenge sampler. We combine these components in Section 6, where we select concrete parameters and evaluate their joint performance. Finally, Section 7 explains how to mitigate the bounded-depth constraint of `PikkuFold` and how to embed the protocol into a more expressive folding scheme supporting AIR relations.

2 Preliminaries

Definition 1 (Indexing and Sampling Conventions). Let $\mathbb{N} = \{1, 2, \dots\}$. For $n \in \mathbb{N}$, define $[n] := \{0, \dots, n-1\}$, and, for $0 \leq m \leq n$, define $[m : n] := \{m, \dots, n-1\}$. For

$i \in [2^\nu]$, write $\text{bin}(i) = (i_0, \dots, i_{\nu-1}) \in \{0, 1\}^\nu$ for its little-endian binary expansion, so that $i = \sum_{j \in [\nu]} 2^j i_j$. The notation $x \leftarrow \$ S$ denotes uniform sampling from a finite set S .

Definition 2 (Balanced Representatives). For an integer $q \geq 2$, write

$$[\pm q/2] := \{-\lceil q/2 \rceil + 1, \dots, \lfloor q/2 \rfloor\}.$$

We identify every element of $\mathbb{Z}_q$ with its unique representative in $[\pm q/2]$. Reduction modulo q of an integer vector or matrix always returns the coordinate-wise balanced representative.

Definition 3 (Linear-Algebra and Computational Conventions). Bold lower-case letters denote vectors and bold upper-case letters denote matrices. The superscript T denotes transpose, and $\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^\mathsf{T} \mathbf{y}$ is the standard bilinear inner product. For vectors or matrices of compatible dimensions, the tuple notation $(\mathbf{M}_i)_{i \in [k]}$ also denotes their horizontal concatenation when this meaning is clear. All logarithms are base two. The security parameter is $\lambda \in \mathbb{N}$, and public parameters implicitly contain 1^λ .

2.1 Cyclotomic Rings and Norms

Definition 4 (Cyclotomic Setting and Embeddings). Let $\mathcal{K} = \mathbb{Q}(\zeta_f)$ be a cyclotomic field of conductor f satisfying $f \bmod 4 \neq 2$, degree $\varphi := \varphi(f)$, and ring of integers $\mathcal{R} := \mathcal{O}_{\mathcal{K}}$. Let q be a prime not dividing f , and write $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ and $\mathcal{R}_q^\times$ for its group of units. We fix the powerful $\mathbb{Z}$ -basis $(b_i)_{i \in [\varphi]}$ of $\mathcal{R}$. For a prime-power conductor this is the power basis, while for a general conductor it is the tensor product of the power bases associated with its pairwise coprime prime-power factors [LPR13].

For $x = \sum_{i \in [\varphi]} x_i b_i \in \mathcal{K}$, define the coefficient embedding $\text{coeff}(x) := (x_i)_{i \in [\varphi]} \in \mathbb{Q}^\varphi$. Let $(\sigma_i)_{i \in [\varphi]}$ be the embeddings of $\mathcal{K}$ into $\mathbb{C}$, and define the canonical embedding $\sigma(x) := (\sigma_i(x))_{i \in [\varphi]} \in \mathbb{C}^\varphi$. Let $i \mapsto \bar{i}$ be the permutation of $[\varphi]$ determined by $\sigma_{\bar{i}}(x) = \overline{\sigma_i(x)}$ for every $x \in \mathcal{K}$, and define the canonical real tensor space

$$H := \{(z_i)_{i \in [\varphi]} \in \mathbb{C}^\varphi : z_{\bar{i}} = \overline{z_i} \text{ for every } i \in [\varphi]\}.$$

Both embeddings extend to vectors by concatenation. For a matrix $\mathbf{X} = (\mathbf{x}_j)_{j \in [c]} \in \mathcal{K}^{b \times c}$, their column-wise extensions are $\text{coeff}(\mathbf{X}) := (\text{coeff}(\mathbf{x}_j))_{j \in [c]} \in \mathbb{Q}^{\varphi b \times c}$ and $\sigma(\mathbf{X}) := (\sigma(\mathbf{x}_j))_{j \in [c]} \in \mathbb{C}^{\varphi b \times c}$. For an object over $\mathcal{R}_q$, coeff first takes the unique lift whose coefficient vector is balanced coordinate-wise. Its canonical embedding and all its norms are defined using the same lift.

Definition 5 (Coefficient and Canonical Norms). For $x \in \mathcal{K}$, define

$$\begin{aligned} \|x\| &:= \|\text{coeff}(x)\|_2, & \|x\|_{\text{coeff}, \infty} &:= \|\text{coeff}(x)\|_\infty, \\ \|x\|_{\sigma, 2} &:= \|\sigma(x)\|_2, & \|x\|_{\sigma, \infty} &:= \|\sigma(x)\|_\infty. \end{aligned}$$

The same notation applies to vectors by concatenating the embeddings of their entries. Thus, the undecorated norm is always the coefficient ℓ_2 -norm. For vectors over $\mathbb{R}$ or $\mathbb{C}$, the undecorated norm is the usual Euclidean norm.

Lemma 1 (Relations between Cyclotomic Norms [KLNO24]). *Define*

$$\widehat{f} := \begin{cases} f, & \text{if } f \text{ is odd,} \\ f/2, & \text{if } f \text{ is even.} \end{cases}$$

For every $m \in \mathbb{N}$ and $\mathbf{x} \in \mathcal{K}^m$,

$$\begin{aligned}\|\mathbf{x}\| &\leq \sqrt{\frac{\text{rad}(\mathfrak{f})}{\mathfrak{f}}} \|\mathbf{x}\|_{\sigma,2}, \\ \|\mathbf{x}\|_{\sigma,2} &\leq \sqrt{\widehat{\mathfrak{f}}} \|\mathbf{x}\|, \\ \|\mathbf{x}\|_{\sigma,\infty} &\leq \varphi \|\mathbf{x}\|_{\text{coeff},\infty}, \\ \|\mathbf{x}\|_{\text{coeff},\infty} &\leq \mathfrak{c}_{\text{rad}(\mathfrak{f})} \|\mathbf{x}\|_{\sigma,\infty}.\end{aligned}$$

Here $\text{rad}(\mathfrak{f})$ is the product of the distinct prime divisors of $\mathfrak{f}$, and $\mathfrak{c}_{\text{rad}(\mathfrak{f})} \leq (4/\pi)^s$, where s is the number of distinct odd prime divisors of $\mathfrak{f}$.

Definition 6 (Operator Norms). For a real or complex matrix $\mathbf{A}$, let $\|\mathbf{A}\|_{\text{op}}$ be its induced Euclidean operator norm. For $c \in \mathcal{K}$, define its coefficient- ℓ_2 multiplication norm by

$$\|c\|_{\text{op}} := \sup_{x \in \mathcal{K} \setminus \{0\}} \frac{\|cx\|}{\|x\|}.$$

Homogeneity implies that the same supremum is obtained over $x \in \mathcal{R} \setminus \{0\}$.

Lemma 2 (Operator Norm). For every $c \in \mathcal{K}$,

$$\sup_{x \in \mathcal{R} \setminus \{0\}} \frac{\|cx\|_{\sigma,2}}{\|x\|_{\sigma,2}} = \|c\|_{\sigma,\infty}.$$

Moreover,

$$\sqrt{\frac{\mathfrak{f}}{\widehat{\mathfrak{f}}\text{rad}(\mathfrak{f})}} \|c\|_{\sigma,\infty} \leq \|c\|_{\text{op}} \leq \sqrt{\frac{\widehat{\mathfrak{f}}\text{rad}(\mathfrak{f})}{\mathfrak{f}}} \|c\|_{\sigma,\infty}.$$

If $\mathfrak{f}$ is a power of two, then $\|c\|_{\text{op}} = \|c\|_{\sigma,\infty}$.

Proof. Under the canonical embedding, multiplication by c is represented by $\text{diag}(\sigma(c))$. Its induced Euclidean operator norm on H is $\|c\|_{\sigma,\infty}$. The image $\sigma(\mathcal{K})$ is dense in H , so restricting the supremum to $\mathcal{K} \setminus \{0\}$ does not change it. Every element of $\mathcal{K}$ has a non-zero integer multiple in $\mathcal{R}$, and the quotient is homogeneous, so the same holds when the supremum is restricted to $\mathcal{R} \setminus \{0\}$.

For every non-zero $x \in \mathcal{K}$, Lemma 1 gives

$$\|cx\| \leq \sqrt{\frac{\text{rad}(\mathfrak{f})}{\mathfrak{f}}} \|c\|_{\sigma,\infty} \|x\|_{\sigma,2} \leq \sqrt{\frac{\widehat{\mathfrak{f}}\text{rad}(\mathfrak{f})}{\mathfrak{f}}} \|c\|_{\sigma,\infty} \|x\|.$$

Taking the supremum proves the upper bound. Applying the same two norm conversions in the opposite direction yields

$$\|c\|_{\sigma,\infty} \leq \sqrt{\frac{\widehat{\mathfrak{f}}\text{rad}(\mathfrak{f})}{\mathfrak{f}}} \|c\|_{\text{op}},$$

which proves the lower bound. For a power-of-two conductor, $\widehat{\mathfrak{f}}\text{rad}(\mathfrak{f})/\mathfrak{f} = 1$. $\square$

2.2 Trace and Ring Lifts

Definition 7 (Trace). The field trace is

$$\text{Tr}_{\mathcal{K}/\mathbb{Q}}(x) := \sum_{i \in [\varphi]} \sigma_i(x).$$

We write $\text{Tr} := \text{Tr}_{\mathcal{K}/\mathbb{Q}}$ and apply it entry-wise to vectors and matrices. Since $\text{Tr}(\mathcal{R}) \subseteq \mathbb{Z}$, it descends to a $\mathbb{Z}_q$ -linear map $\text{Tr}: \mathcal{R}_q \rightarrow \mathbb{Z}_q$.

Lemma 3 (Trace-Dual Basis [KLNO25, Lemma 1]). *There are elements $(b_i^\vee)_{i \in [\varphi]}$ in $\mathcal{R}$ such that*

$$\mathrm{Tr}(b_i b_j^\vee) = \delta_{i,j} \bmod q \quad \text{for every } i, j \in [\varphi].$$

Definition 8 (Fine and Coarse Ring Lifts). Fix a trace-dual basis as in Lemma 3. Define

$$\mathrm{coeff}_\vee^{-1}(z_0, \dots, z_{\varphi-1}) := \sum_{i \in [\varphi]} z_i b_i^\vee$$

and extend this map blockwise. For $\mathbf{J} \in \mathbb{Z}_q^{a \times \varphi b}$, define its fine lift $\mathrm{Lift}_{\mathrm{fine}}(\mathbf{J}) \in \mathcal{R}_q^{a \times b}$ by

$$(\mathrm{Lift}_{\mathrm{fine}}(\mathbf{J}))_{s,t} := \sum_{i \in [\varphi]} J_{s, \varphi t + i} b_i^\vee.$$

For $\mathbf{J} \in \mathbb{Z}_q^{a \times b}$, define its coarse lift $\mathrm{Lift}_{\mathrm{coarse}}(\mathbf{J}) \in \mathcal{R}_q^{a \times b}$ by applying the natural embedding $\mathbb{Z}_q \rightarrow \mathcal{R}_q$ entry-wise.

Lemma 4 (Lift Identities). *For every $\mathbf{z} \in \mathbb{Z}_q^{\varphi b}$ and $\mathbf{x} \in \mathcal{R}_q^b$,*

$$\mathrm{Tr}(\langle \mathrm{coeff}_\vee^{-1}(\mathbf{z}), \mathbf{x} \rangle) = \mathbf{z}^\top \mathrm{coeff}(\mathbf{x}) \bmod q.$$

Consequently, for every $\mathbf{W} \in \mathcal{R}_q^{b \times c}$ and matrices $\mathbf{J}$ of the dimensions required by Definition 8,

$$\mathrm{Tr}(\mathrm{Lift}_{\mathrm{fine}}(\mathbf{J})\mathbf{W}) = \mathbf{J} \mathrm{coeff}(\mathbf{W}) \bmod q, \quad \mathrm{Tr}(\mathrm{Lift}_{\mathrm{coarse}}(\mathbf{J})\mathbf{W}) = \mathbf{J} \mathrm{Tr}(\mathbf{W}) \bmod q.$$

Moreover, for every $\mathbf{w} \in \mathcal{R}_q^b$,

$$\mathrm{coeff}(\mathrm{Lift}_{\mathrm{coarse}}(\mathbf{J})\mathbf{w}) = (\mathbf{J} \otimes \mathbf{I}_\varphi) \mathrm{coeff}(\mathbf{w}) \bmod q.$$

Proof. Expand the inner product in the powerful basis and apply $\mathrm{Tr}(b_i b_j^\vee) = \delta_{i,j} \bmod q$. This proves the first identity. Applying it to every entry of the matrix product proves the fine-lift identity. The remaining identities follow from the $\mathbb{Z}_q$ -linearity of the trace and coefficient maps. $\square$

2.3 Residue-Ring Decomposition and Batching

Definition 9 (NTT Decomposition). Suppose that the cyclotomic polynomial Φ_f factors modulo q into φ/e distinct irreducible factors of degree e . Fix the induced ring isomorphism

$$\mathrm{NTT}: \mathcal{R}_q \longrightarrow (\mathbb{F}_{q^e})^{\varphi/e}.$$

For every $a \mid e$, fix the subfield $\mathbb{F}_{q^a} \subseteq \mathbb{F}_{q^e}$ and an $\mathbb{F}_{q^a}$ -basis of $\mathbb{F}_{q^e}$. As an $\mathbb{F}_{q^a}$ -module, $\mathbb{F}_{q^e}$ has rank e/a . Expanding every NTT slot in the fixed basis therefore defines an $\mathbb{F}_{q^a}$ -linear isomorphism

$$\theta_a: \mathcal{R}_q \longrightarrow (\mathbb{F}_{q^a})^{\varphi/a}.$$

Definition 10 (Subfield Batching). Fix $a \mid e$ for e from Definition 9. For $\boldsymbol{\delta} \leftarrow \$ (\mathbb{F}_{q^a})^{\varphi/a}$, define the $\mathbb{F}_{q^a}$ -linear map

$$\Phi_{\boldsymbol{\delta}}: \mathcal{R}_q \longrightarrow \mathbb{F}_{q^a}, \quad \Phi_{\boldsymbol{\delta}}(x) := \boldsymbol{\delta}^\top \theta_a(x).$$

The map acts coefficient-wise on polynomials over $\mathcal{R}_q$. For every

$$g \in \mathcal{R}_q[x_0, \dots, x_{\ell-1}] \quad \text{and} \quad \mathbf{r} \in (\mathbb{F}_{q^a})^\ell,$$

we evaluate $g(\mathbf{r})$ in each NTT slot using the inclusion $\mathbb{F}_{q^a} \subseteq \mathbb{F}_{q^e}$. Linearity then gives

$$\Phi_{\boldsymbol{\delta}}(g)(\mathbf{r}) = \Phi_{\boldsymbol{\delta}}(g(\mathbf{r})).$$

Lemma 5 (Subfield-Batching Error). *For every non-zero $x \in \mathcal{R}_q$,*

$$\Pr_{\delta \leftarrow \mathbb{S}(\mathbb{F}_{q^a})^{\varphi/a}} [\Phi_{\delta}(x) = 0] = q^{-a}.$$

Proof. The vector $\theta_a(x)$ is non-zero because θ_a is an isomorphism. Fix a non-zero coordinate. Conditional on every other coordinate of δ , exactly one value of the corresponding batching coefficient makes the inner product vanish. $\square$

2.4 Challenge Sets and Coordinate-Wise Extraction

Definition 11 (Strong and Approximate Strong Sampling Sets). Let $\mathcal{C} \subseteq \mathcal{R}$ be finite. It is a strong sampling set modulo q if $c - c'$ is a unit in $\mathcal{R}_q$ for every distinct $c, c' \in \mathcal{C}$. It is an $\epsilon_{\mathcal{C}}$ -approximate strong sampling set modulo q if

$$\max_{c_0 \in \mathcal{C}} \Pr_{c \leftarrow \mathbb{S}\mathcal{C}} [c \neq c_0 \wedge c - c_0 \notin \mathcal{R}_q^{\times}] \leq \epsilon_{\mathcal{C}}.$$

Its coefficient- ℓ_2 expansion factor is

$$\Gamma_{\mathcal{C}} := \max_{c \in \mathcal{C}} \|c\|_{\text{op}}.$$

Thus, strong sampling is the special case $\epsilon_{\mathcal{C}} = 0$.

Remark 1. This definition is strictly stronger than that of [GLLO26] for the same ϵ . In that definition, both challenges c and c' are sampled independently from $\mathcal{C}$. We believe that this is a more natural notion in the context of coordinate-wise forking, where one (base) transcript is fixed and the remaining transcripts are sampled via rewinding.

Definition 12 (Coordinate-Wise Special-Soundness Families [FMN24]). Let $k, \ell \in \mathbb{N}$, let S be a finite set, and define $K := \ell(k - 1) + 1$. The family $\text{SS}(S, \ell, k)$ consists of the sets $C = \{\mathbf{c}_0, \dots, \mathbf{c}_{K-1}\} \subseteq S^{\ell}$ of cardinality K for which there is a central index $e \in [K]$ such that, for every $i \in [\ell]$, there is a set $J_i \subseteq [K] \setminus \{e\}$ of cardinality $k - 1$ satisfying, for every $j \in J_i$,

$$c_{e,i} \neq c_{j,i} \quad \text{and} \quad c_{e,h} = c_{j,h} \quad \text{for every } h \in [\ell] \setminus \{i\}.$$

Let Ξ be a relation. A public-coin challenge-response stage with challenge space S^{ℓ} is ℓ -coordinate-wise k -special sound for Ξ if a polynomial-time algorithm extracts a witness for Ξ from K accepting transcripts with a common instance and prefix and a challenge set in $\text{SS}(S, \ell, k)$.

Lemma 6 (Coordinate-Wise Forking [FMN24, Lemma 7.1]). *Let $k, \ell \in \mathbb{N}$, let S be a finite set of cardinality N , define $\mathcal{C} := S^{\ell}$, and let $V: \mathcal{C} \times \{0, 1\}^* \rightarrow \{0, 1\}$ be any verification function. There is an oracle algorithm $\mathcal{E}$ with the following properties. Given oracle access to a probabilistic algorithm $\mathcal{A}: \mathcal{C} \rightarrow \{0, 1\}^*$, the algorithm $\mathcal{E}^{\mathcal{A}}$ makes at most $\ell(k - 1) + 1$ queries to $\mathcal{A}$ in expectation and, with probability at least*

$$\epsilon_V(\mathcal{A}) - \frac{\ell(k - 1)}{N}, \quad \epsilon_V(\mathcal{A}) := \Pr_{\mathbf{c} \leftarrow \mathbb{S}\mathcal{C}} [V(\mathbf{c}, \mathcal{A}(\mathbf{c})) = 1],$$

outputs $\ell(k - 1) + 1$ pairs $(\mathbf{c}_i, y_i)_{i \in [\ell(k - 1) + 1]}$ such that $V(\mathbf{c}_i, y_i) = 1$ for every $i \in [\ell(k - 1) + 1]$ and

$$\{\mathbf{c}_i : i \in [\ell(k - 1) + 1]\} \in \text{SS}(S, \ell, k).$$

2.5 Polynomial Tools

Lemma 7 (Polynomial Root Bound [BCPS18, Theorem 4.2]). *Let $\mathbb{F}$ be a field, let $S \subseteq \mathbb{F}$ be finite, and let $f \in \mathbb{F}[x_0, \dots, x_{\ell-1}]$ be a non-zero polynomial of total degree at most D . Then*

$$\Pr_{\mathbf{r} \leftarrow \mathbb{S} S^\ell} [f(\mathbf{r}) = 0] \leq \frac{D}{|S|}.$$

Definition 13 (Equality Polynomial). For $\mathbf{b} \in \{0, 1\}^\nu$ and $\mathbf{x} \in \mathcal{R}_q^\nu$, define

$$\text{eq}(\mathbf{b}, \mathbf{x}) := \prod_{j \in [\nu]} (b_j x_j + (1 - b_j)(1 - x_j)).$$

Its expanded tensor vector is

$$\tilde{\text{eq}}(\mathbf{x}) := \bigotimes_{j \in [\nu]} (1 - x_j, x_j) = (\text{eq}(\mathbf{b}, \mathbf{x}))_{\mathbf{b} \in \{0, 1\}^\nu} \in \mathcal{R}_q^{2^\nu},$$

where Boolean vectors are ordered lexicographically. On the image of $\tilde{\text{eq}}$, define

$$(\tilde{\text{eq}}^{-1}(\mathbf{u}))_j := \sum_{\substack{\mathbf{b} \in \{0, 1\}^\nu \\ b_j = 1}} u_{\mathbf{b}}.$$

Then $\tilde{\text{eq}}^{-1}(\tilde{\text{eq}}(\mathbf{x})) = \mathbf{x}$. Indeed, the j th coordinate of the left-hand side is $x_j \prod_{h \in [\nu] \setminus \{j\}} ((1 - x_h) + x_h) = x_j$.

Definition 14 (Multilinear Extensions). For $\mathbf{w} \in \mathcal{R}_q^{2^\nu}$, define

$$\text{MLE}[\mathbf{w}](\mathbf{x}) := \sum_{\mathbf{b} \in \{0, 1\}^\nu} w_{\mathbf{b}} \text{eq}(\mathbf{b}, \mathbf{x}) = \tilde{\text{eq}}(\mathbf{x})^\top \mathbf{w}.$$

For $\mathbf{A} \in \mathcal{R}_q^{2^{\nu'} \times 2^\nu}$, define

$$\text{MLE}[\mathbf{A}](\mathbf{x}', \mathbf{x}) := \sum_{\substack{\mathbf{b}' \in \{0, 1\}^{\nu'} \\ \mathbf{b} \in \{0, 1\}^\nu}} A_{\mathbf{b}', \mathbf{b}} \text{eq}(\mathbf{b}', \mathbf{x}') \text{eq}(\mathbf{b}, \mathbf{x}).$$

Definition 15 (Ring Sumcheck). Fix $a \mid e$ and the maps in Definitions 9 and 10. Let $g \in \mathcal{R}_q[x_0, \dots, x_{\ell-1}]$ have individual degree at most D , and let $t \in \mathcal{R}_q$. To verify

$$\sum_{\mathbf{b} \in \{0, 1\}^\ell} g(\mathbf{b}) = t,$$

the protocol RingSC samples $\boldsymbol{\delta} \leftarrow \mathbb{S} (\mathbb{F}_{q^a})^{\varphi/a}$ and runs the standard sumcheck protocol over $\mathbb{F}_{q^a}$ on

$$G := \Phi_{\boldsymbol{\delta}}(g), \quad T := \Phi_{\boldsymbol{\delta}}(t).$$

The terminal point $\mathbf{r}$ is uniform in $(\mathbb{F}_{q^a})^\ell$. At this point, the prover sends $g(\mathbf{r}) \in \mathcal{R}_q$, and the verifier checks that its subfield batching equals the terminal value claimed for $G(\mathbf{r})$. The application supplies the final check of $g(\mathbf{r})$.

Lemma 8 (Ring-Sumcheck Soundness Error, Adapted from [KLN⁺26, Section 7]). *The protocol RingSC is perfectly complete. If the terminal evaluation is checked correctly, it accepts a false ring claim with probability at most*

$$\kappa_{\text{sc}}(\ell, D) := \frac{\ell D + 1}{q^a}.$$

When each round polynomial omits the coefficient fixed by its consistency equation, prover communication is $D\ell$ elements of $\mathbb{F}_{q^a}$ and one element of $\mathcal{R}_q$. If all verifier randomness is sent explicitly, verifier communication is $\varphi/a + \ell$ elements of $\mathbb{F}_{q^a}$.

Proof. If the ring claim is false, its discrepancy is a non-zero element of $\mathcal{R}_q$. By Lemma 5, subfield batching maps this discrepancy to zero with probability q^{-a} . Outside this event, the resulting field claim is false. Standard sumcheck soundness contributes at most $\ell D/q^a$ [LFKN92]. The union bound proves the soundness claim. In each of the ℓ rounds, the prover sends a polynomial of degree at most D . Its value-sum constraint fixes one coefficient, so D field elements determine it. The terminal ring evaluation contributes one element of $\mathcal{R}_q$. The verifier sends the φ/a batching coefficients and one field challenge in each sumcheck round. $\square$

2.6 Johnson–Lindenstrauss Projections

Definition 16 (Euclidean Projection Guarantee). Let χ be a distribution over $\mathbb{Z}$, and let $\kappa \in (0, 1)$. Parameters $n_{\text{rp}}(\kappa)$, $\alpha_{\text{rp}}(\kappa)$, $\beta_{\text{rp}}(\kappa)$, and $b(\kappa)$ satisfy the Euclidean projection guarantee for χ if the following statements hold for every $m \in \mathbb{N}$ and $\mathbf{J} \leftarrow_{\$} \chi^{n_{\text{rp}}(\kappa) \times m}$.

1. For every non-zero $\mathbf{w} \in \mathbb{R}^m$, in particular for every non-zero integer vector,

$$\Pr \left[\frac{\|\mathbf{J}\mathbf{w}\|}{\|\mathbf{w}\|} \notin [\alpha_{\text{rp}}(\kappa), \beta_{\text{rp}}(\kappa)] \right] \leq \kappa.$$

2. For every integer $q \geq 2$, every $0 < \theta \leq q/b(\kappa)$, and every $\mathbf{w} \in [\pm q/2]^m$ satisfying $\|\mathbf{w}\| \geq \theta$,

$$\Pr [\|\mathbf{J}\mathbf{w} \bmod q\| \leq \alpha_{\text{rp}}(\kappa)\theta] \leq \kappa.$$

The parameter $b(\kappa)$ is the wrap-around margin. Item 2 certifies norm thresholds only up to $q/b(\kappa)$ and thereby quantifies how far below the modulus the threshold must lie for the modular lower-tail guarantee.

Remark 2 (Concrete Projection Constants). For the sparse ternary distribution with probabilities $1/4$, $1/2$, and $1/4$ on -1 , 0 , and 1 , respectively, the 128-bit parameters of [BS23, Lemmas 4.1 and 4.2] are $n_{\text{rp}} = 256$, $\alpha_{\text{rp}} = \sqrt{30}$, $\beta_{\text{rp}} = \sqrt{337}$, and $b = 125$, chosen heuristically. Theorem 2 gives a self-contained parameterised analysis and concrete values without relying on heuristics.

2.7 The SIS Assumption

We state the short integer solution (SIS) problem over $\mathcal{R}$ [Ajt96], in the form used in [OKC⁺25].

Definition 17 (SIS). Let $m, n \in \mathbb{N}$ and $\beta \in \mathbb{R}$. The $\text{SIS}_{\mathcal{R}, q, m, n, \beta}$ problem is, given $\mathbf{F} \leftarrow_{\$} \mathcal{R}_q^{n \times m}$, to find $\mathbf{x} \in \mathcal{R}^m$ with

$$\mathbf{x} \neq 0; \|\mathbf{x}\| \leq \beta; \mathbf{F}\mathbf{x} = 0 \bmod q.$$

Definition 18 (SIS Hardness). For parameters depending on the security parameter λ , the $\text{SIS}_{\mathcal{R}, q, m, n, \beta}$ problem is *hard* if, for every probabilistic expected polynomial-time adversary $\mathcal{A}$,

$$\Pr [\mathbf{x} \neq 0 \wedge \|\mathbf{x}\| \leq \beta \wedge \mathbf{F}\mathbf{x} = 0 \bmod q \mid \mathbf{F} \leftarrow_{\$} \mathcal{R}_q^{n \times m}, \mathbf{x} \leftarrow \mathcal{A}(\mathbf{F})] \leq 2^{-\lambda}.$$

2.8 Reductions of Knowledge and Folding Schemes

Definition 19 (Reduction of Knowledge, Adapted from [KLNO24, OKC⁺25, KLOT25]). A reduction of knowledge (RoK) $\Pi = (\mathcal{P}, \mathcal{V})$ consists of interactive probabilistic polynomial-time algorithms with the following interface:

- $\mathcal{P}(\text{pp}, \text{stmt}_0, \text{wit}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ interactively reduces an input instance to an output instance or returns $\perp$.
- $\mathcal{V}(\text{pp}, \text{stmt}_0) \rightarrow \text{stmt}_1$ interactively reduces verification of the input statement to verification of the output statement.

We write $\langle \mathcal{P}, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0, \text{wit}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ for an honest interaction. The notation $\langle \mathcal{P}^*, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ denotes an interaction with a malicious prover.

Let Ξ_0 and Ξ_1 be relations. The RoK is correct for $\Xi_0 \rightarrow \Xi_1$ with correctness error δ if, for every $(\text{pp}, \text{stmt}_0, \text{wit}_0) \in \Xi_0$,

$$\Pr[(\text{pp}, \text{stmt}_1, \text{wit}_1) \in \Xi_1 \mid (\text{stmt}_1, \text{wit}_1) \leftarrow \langle \mathcal{P}, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0, \text{wit}_0)] \geq 1 - \delta(\text{pp}, \text{stmt}_0).$$

Let Ξ'_0 and Ξ'_1 be extraction relations. The RoK is knowledge sound for $\Xi'_0 \leftarrow \Xi'_1$ with knowledge error κ if there is a black-box expected polynomial-time extractor $\mathcal{E}$ such that, for every $(\text{pp}, \text{stmt}_0)$ and every possibly unbounded malicious prover $\mathcal{P}^*$,

$$\begin{aligned} & \Pr[(\text{pp}, \text{stmt}_0, \text{wit}_0) \in \Xi'_0 \mid \text{wit}_0 \leftarrow \mathcal{E}^{\mathcal{P}^*}(\text{pp}, \text{stmt}_0)] \\ & \geq \Pr[(\text{pp}, \text{stmt}_1, \text{wit}_1) \in \Xi'_1 \mid (\text{stmt}_1, \text{wit}_1) \leftarrow \langle \mathcal{P}^*, \mathcal{V} \rangle(\text{pp}, \text{stmt}_0)] - \kappa(\text{pp}, \text{stmt}_0). \end{aligned}$$

Oracle calls to $\mathcal{P}^*$ have unit cost. The RoK is depth-preserving knowledge sound if $\text{Depth}(\mathcal{E}^{\mathcal{P}^*}) \leq \text{Depth}(\mathcal{P}^*) + \text{poly}(\lambda)$.

Following [KST22], we view a folding scheme as a structure-preserving RoK. It reduces input instances and an accumulator instance to a new accumulator instance while preserving a common structure, which in this work is the commitment key.

Definition 20 (Structure Preservation [KLOT25]). Let $k \in \mathbb{N}$, and let $(\Xi_i)_{i \in [k]}$ and Ξ_{out} be relations. Equip their statements with a structure map str into a common set $\mathcal{S}$. For $s \in \mathcal{S}$, define

$$\Xi_i \upharpoonright_s := \{(\text{pp}, \text{stmt}, \text{wit}) \in \Xi_i : \text{str}(\text{stmt}) = s\},$$

and define $\Xi_{\text{out}} \upharpoonright_s$ analogously. A RoK Π is structure-preserving for $\prod_{i \in [k]} \Xi_i \rightarrow \Xi_{\text{out}}$ with correctness error δ if, for every $s \in \mathcal{S}$, it is correct for

$$\prod_{i \in [k]} (\Xi_i \upharpoonright_s) \longrightarrow \Xi_{\text{out}} \upharpoonright_s$$

with error δ .

Definition 21 (Relaxed Folding Scheme, Adapted from [KLOT25]). Let $k \in \mathbb{N}$. Let $(\Xi_i)_{i \in [k]}$, $\Xi_{\text{acc}, \text{in}}$, $\Xi_{\text{acc}, \text{out}}$, $(\Xi'_i)_{i \in [k]}$, $\Xi'_{\text{acc}, \text{in}}$, $\Xi'_{\text{acc}, \text{out}}$, and Ξ_{break} be relations equipped with structure maps into a common set $\mathcal{S}$. A RoK Π is a relaxed folding scheme with correctness error δ if it is structure-preserving for

$$\left(\prod_{i \in [k]} \Xi_i \right) \times \Xi_{\text{acc}, \text{in}} \longrightarrow \Xi_{\text{acc}, \text{out}}$$

with error δ . It is knowledge sound with error κ if, for every $s \in \mathcal{S}$, it is knowledge sound for

$$\left(\left(\prod_{i \in [k]} (\Xi'_i \upharpoonright_s) \right) \times (\Xi'_{\text{acc}, \text{in}} \upharpoonright_s) \right) \cup (\Xi_{\text{break}} \upharpoonright_s) \longleftarrow \Xi'_{\text{acc}, \text{out}} \upharpoonright_s$$

with error κ . The qualifier “relaxed” permits the input and output accumulator relations to differ.

3 Folding Scheme

Consider a fresh opening $(\mathbf{y}, \mathbf{w})$ and an accumulator opening $(\mathbf{y}_{\text{acc}}, \mathbf{w}_{\text{acc}})$ under the same commitment matrix $\mathbf{F}$. The most direct candidate folding protocol samples a short challenge z and forms

$$\mathbf{w}_{\text{fold}} := \mathbf{w}_{\text{acc}} + z\mathbf{w}, \quad \mathbf{y}_{\text{fold}} := \mathbf{y}_{\text{acc}} + z\mathbf{y} \bmod q.$$

Correctness is immediate, since $\mathbf{F}\mathbf{w}_{\text{fold}} = \mathbf{y}_{\text{fold}} \bmod q$. Moreover, if $\|z\|_{\text{op}} \leq \gamma$, then

$$\|\mathbf{w}_{\text{fold}}\| \leq \|\mathbf{w}_{\text{acc}}\| + \gamma\|\mathbf{w}\|.$$

The accumulator witness is not multiplied by the challenge, so its norm grows only by the contribution of the fresh witness.

The difficulty appears in extraction. Suppose that two accepting executions with the same prefix produce output witnesses $\mathbf{u}_0, \mathbf{u}_1$ for distinct challenges z_0, z_1 . Set

$$\Delta := z_0 - z_1, \quad \mathbf{a} := \mathbf{u}_0 - \mathbf{u}_1.$$

If the output witnesses have norm at most β_{out} and $\|z_i\|_{\text{op}} \leq \gamma$, then

$$\mathbf{F}\mathbf{a} = \Delta\mathbf{y} \bmod q, \quad \|\mathbf{a}\| \leq 2\beta_{\text{out}}, \quad \|\Delta\|_{\text{op}} \leq 2\gamma.$$

Thus, the extractor obtains a short numerator $\mathbf{a}$ and a short denominator Δ . Recovering witnesses for the original fresh and accumulator statements, however, requires division by that denominator:

$$\mathbf{w}^* = \frac{\mathbf{u}_0 - \mathbf{u}_1}{z_0 - z_1} = \Delta^{-1}\mathbf{a} \bmod q, \quad \mathbf{w}_{\text{acc}}^* = \frac{z_0\mathbf{u}_1 - z_1\mathbf{u}_0}{z_0 - z_1} \bmod q.$$

Even when Δ is invertible modulo q , its inverse need not be short. Consequently, the available bounds are only

$$\|\mathbf{w}^*\| \leq 2\beta_{\text{out}}\|\Delta^{-1}\|_{\text{op}}, \quad \|\mathbf{w}_{\text{acc}}^*\| \leq 2\gamma\beta_{\text{out}}\|\Delta^{-1}\|_{\text{op}},$$

and $\|\Delta^{-1}\|_{\text{op}}$ has no useful bound. The candidate witnesses may therefore be far outside the relations' prescribed norm bounds.

This suggests addressing the problem separately for the fresh input. If a separate protocol certifies that the extracted fresh witness $\mathbf{w}^*$ is short, then the accumulator witness can be reconstructed as $\mathbf{u}_0 - z_0\mathbf{w}^*$. Its norm incurs only additive growth, and the accumulator itself requires no norm check.

Before describing our solution, we formalise the relation that the folding protocol must satisfy. It is a principal relation in the sense of [KLNO24]: the witness is a short vector $\mathbf{w}$ committed under $\mathbf{F}$. To make the relation useful downstream, it additionally carries a multilinear-evaluation claim on $\mathbf{w}$. The matrix $\mathbf{F}$ is to be viewed as a commitment key.

Definition 22 (Principal Relation). Let $m, n \in \mathbb{N}$, $\beta \in \mathbb{R}$. We define the relation

$$\Xi_{\mathcal{R}, q, m, n, \beta}^{\text{lin}} := \left\{ ((\mathbf{F}, \mathbf{y}, \mathbf{s}, t), \mathbf{w}) : \begin{array}{l} \mathbf{F} \in \mathcal{R}_q^{n \times m}; \mathbf{y} \in \mathcal{R}_q^n; \mathbf{w} \in \mathcal{R}^m \\ \mathbf{s} \in \mathcal{R}_q^{\log_2 m}; t \in \mathcal{R}_q \\ \text{MLE}[\mathbf{w}](\mathbf{s}) = t \bmod q \\ \|\mathbf{w}\| \leq \beta \quad \mathbf{F}\mathbf{w} = \mathbf{y} \bmod q \end{array} \right\}$$

A standard approach to proving that a witness $\mathbf{w}$ is short is to send a Johnson–Lindenstrauss (JL) projection of it. A JL projection gives the desired norm certificate. For suitable factors $\ell, u > 0$, a sampled matrix $\mathbf{J}$ satisfies

$$\ell \|\mathbf{w}\| \leq \|\mathbf{J}\mathbf{w}\| \leq u \|\mathbf{w}\|$$

with high probability. The prover sends the projection image $\mathbf{v} = \mathbf{J}\mathbf{w}$, and the verifier checks that $\mathbf{v}$ is short. Proving $\mathbf{J}\mathbf{w} = \mathbf{v}$ then allows the extractor to infer that $\mathbf{w}$ is short as well.

The first natural implementation, used in the lattice-based proof systems of [BS23, NS24], uses an unstructured JL matrix $\mathbf{J}$. Its projection image is small enough to send directly, but verification of $\mathbf{J}\mathbf{w} = \mathbf{v}$ is the obstruction. Although this equality is relatively easy to prove, the verifier must evaluate an unstructured matrix whose width is the full witness dimension. Its running time is therefore linear in the witness size.

A second natural attempt is to use the structured projection introduced in [KLNO25] and incorporated into the folding schemes of [KLOT25, Osa26]. Instead of a dense matrix, it uses a block projection $\mathbf{I}_r \otimes \mathbf{J}_0$, where the small matrix $\mathbf{J}_0 \in \mathbb{Z}^{n_0 \times m_0}$ is shared across all blocks. Its description and multilinear extension are succinct. Writing $\varsigma := m_0/n_0$ for the shrinkage, however, one layer maps a witness of dimension N_0 only to an image of dimension N_0/ς , which is still too large to send. Prior work therefore commits to this image, adding an entire commitment to the communication transcript.

Our solution is a *layered random projection* (LRP). We recursively project the output of each layer into the next layer until the image is short. Every layer has a succinct description and reduces the current dimension, while preserving the norm up to controlled factors. The intermediate projection images remain local to the prover. Only the final image is sent to the verifier, after it has become short enough for direct transmission. The projection structure may vary between layers. In our instantiation, the earlier layers project complete ring elements, while the final layer descends to their coefficients. Figure 1 compares the three approaches.

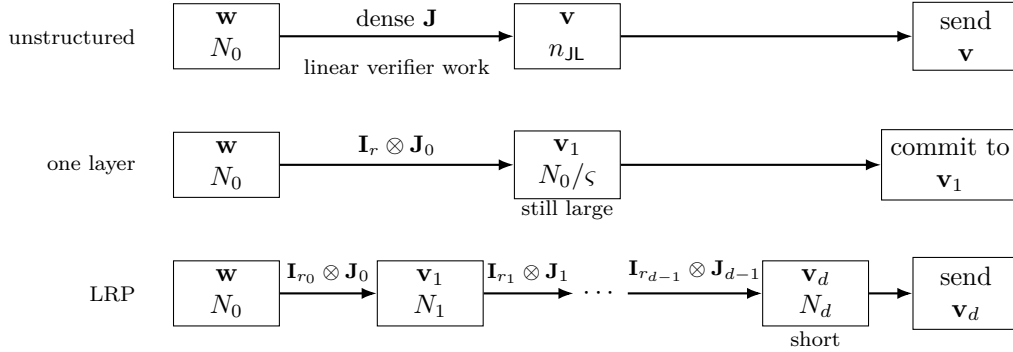


Figure 1: Projection strategies for a JL norm certificate.

It remains to verify the projection without revealing its intermediate images or forming the dense product of all layer matrices. This relation is naturally amenable to sumcheck. Matrix multiplication already sums over the coordinates between consecutive layers. We retain these intermediate coordinates as the summation variables and use the multilinear extensions of the individual layer matrices as factors. A single ring sumcheck then verifies the factorised matrix–vector product. At the terminal point, the verifier evaluates each succinct layer separately and never materialises their product.

Furthermore, we use the ring sumcheck to “homogenise” the evaluation claims. By Definition 22, the fresh and accumulator relations each carry an evaluation claim on their

witness, generally at different points $\mathbf{s}$ and $\mathbf{s}_{\text{acc}}$. These claims cannot be folded directly, since the multilinear extension of the folded witness must be evaluated at one point. We first rewrite them as the sumcheck identities

$$\begin{aligned} t &= \sum_{\mathbf{b} \in \{0,1\}^{\log_2 m}} \text{MLE}[\mathbf{w}](\mathbf{b}) \text{eq}(\mathbf{b}, \mathbf{s}) \pmod{q}, \\ t_{\text{acc}} &= \sum_{\mathbf{b} \in \{0,1\}^{\log_2 m}} \text{MLE}[\mathbf{w}_{\text{acc}}](\mathbf{b}) \text{eq}(\mathbf{b}, \mathbf{s}_{\text{acc}}) \pmod{q}. \end{aligned}$$

We then batch these identities with the sumcheck for the JL projection. Since all batched claims use the same sumcheck challenges, the only witness-dependent claims left at the terminal point are

$$t' = \text{MLE}[\mathbf{w}](\mathbf{r}), \quad t'_{\text{acc}} = \text{MLE}[\mathbf{w}_{\text{acc}}](\mathbf{r})$$

at the same random point $\mathbf{r}$. They can now be folded by linearity:

$$\text{MLE}[\mathbf{w}_{\text{acc}} + z\mathbf{w}](\mathbf{r}) = t'_{\text{acc}} + zt' \pmod{q}.$$

The remainder of this section formalises this intuition. Section 3.1 proves the norm-preserving properties of layered random projections, Section 3.2 describes the folding protocol, and Section 3.3 analyses its efficiency.

3.1 Layered Random Projections

Lemma 9 (Layered random projections). *Let $d, q \in \mathbb{N}$, let $(r_i)_{i \in [d]}$ be positive integers, and let $(m_i, n_i, \kappa_i)_{i \in [d]}$ satisfy $m_i, n_i \in \mathbb{N}$, $\kappa_i \in (0, 1)$, and*

$$r_i m_i = r_{i-1} n_{i-1} \quad \text{for every } i \in [1 : d].$$

Assume that every n_i is chosen so that the guarantee of Definition 16 holds for matrices sampled from $\chi^{n_i \times m_i}$ with error κ_i . For $i \in [d]$, let $\ell_i := \alpha_{\text{rp}}(\kappa_i)$, $u_i := \beta_{\text{rp}}(\kappa_i)$, and $b_i := b(\kappa_i)$, and sample the matrices $(\mathbf{J}_i)_{i \in [d]}$ independently with

$$\mathbf{J}_i \leftarrow \chi^{n_i \times m_i}.$$

Define

$$\mathbf{P} := (\mathbf{I}_{r_{d-1}} \otimes \mathbf{J}_{d-1}) \cdots (\mathbf{I}_{r_1} \otimes \mathbf{J}_1)(\mathbf{I}_{r_0} \otimes \mathbf{J}_0).$$

Then, for every non-zero $\mathbf{w} \in \mathbb{Z}^{r_0 m_0}$,

$$\Pr \left[\frac{\|\mathbf{P}\mathbf{w}\|}{\|\mathbf{w}\|} \notin \left[\prod_{i \in [d]} \ell_i, \prod_{i \in [d]} u_i \right] \right] \leq \sum_{i \in [d]} r_i \kappa_i.$$

Moreover, for every $\theta > 0$ satisfying

$$\theta \prod_{j \in [i]} \ell_j \leq \frac{q}{b_i} \quad \text{for every } i \in [d],$$

and every $\mathbf{w} \in [\pm q/2]^{r_0 m_0}$ with $\|\mathbf{w}\| \geq \theta$, we have

$$\Pr \left[\|\mathbf{P}\mathbf{w} \bmod q\| \leq \theta \prod_{i \in [d]} \ell_i \right] \leq \sum_{i \in [d]} r_i \kappa_i.$$

In particular, setting $\kappa_i := \kappa/(dr_i)$ gives failure probability at most κ .

Proof. For $i \in [d]$, process the layers in the order $0, \dots, d-1$ and decompose the input to $\mathbf{I}_{r_i} \otimes \mathbf{J}_i$ into its r_i blocks. Conditional on the preceding layers, these blocks are fixed. By Definition 16, the norm of every non-zero block is preserved within $[\ell_i, u_i]$ except with probability at most κ_i . A union bound over the r_i blocks shows that the i th layer preserves the norm of its input within $[\ell_i, u_i]$ except with probability at most $r_i \kappa_i$. A second union bound over the d layers gives the stated failure probability. When every layer succeeds, the per-layer lower and upper bounds multiply.

For the modular claim, process the layers in the order $0, \dots, d-1$ and set $\theta_i := \theta \prod_{j \in [i]} \ell_j$, so that $\theta_i \leq q/b_i$. At the input of the i th layer, reduce every coordinate modulo q to its balanced representative and partition the resulting vector into its r_i blocks. Conditional on the preceding layers, suppose that its norm is at least θ_i . If every block has norm at most q/b_i , the modular part of Definition 16, applied to each non-zero block with threshold equal to its norm, shows except with probability at most $r_i \kappa_i$ that the output norm is greater than ℓ_i times the input norm. Otherwise, applying the same guarantee to any block of norm greater than q/b_i , with threshold q/b_i , shows except with probability at most κ_i that the output norm is greater than $\ell_i q/b_i \geq \ell_i \theta_i$. Thus, except with probability at most $r_i \kappa_i$, the i th modular layer maps an input of norm at least θ_i to an output of norm greater than $\ell_i \theta_i$. A union bound over the layers proves the claim, since reducing modulo q after every layer gives the same final vector as reducing $\mathbf{P}\mathbf{w}$ modulo q . $\square$

Lemma 10 (Layered fine/coarse ring embedding). *Let $d \in \mathbb{N}$, let q be an unramified prime for $\mathcal{R}$, and let $(r_i, m_i, n_i)_{i \in [d]}$ be positive integers with $r_{d-1} = 1$ and*

$$r_i n_i = r_{i+1} m_{i+1} \quad \text{for every } i \in [d-1].$$

For arbitrary matrices

$$\mathbf{J}_i \in \mathbb{Z}_q^{n_i \times m_i} \quad \text{for } i \in [d-1], \quad \mathbf{J}_{d-1} \in \mathbb{Z}_q^{n_{d-1} \times (\varphi m_{d-1})},$$

set

$$\mathbf{M}_i := \text{Lift}_{\text{coarse}}(\mathbf{J}_i) \in \mathcal{R}_q^{n_i \times m_i} \quad \text{for } i \in [d-1], \quad \mathbf{M}_{d-1} := \text{Lift}_{\text{fine}}(\mathbf{J}_{d-1}) \in \mathcal{R}_q^{n_{d-1} \times m_{d-1}},$$

and

$$\mathbf{M} := (\mathbf{I}_{r_{d-1}} \otimes \mathbf{M}_{d-1}) \cdots (\mathbf{I}_{r_1} \otimes \mathbf{M}_1)(\mathbf{I}_{r_0} \otimes \mathbf{M}_0) \in \mathcal{R}_q^{n_{d-1} \times r_0 m_0}.$$

For $i \in [d-1]$, define

$$\mathbf{C}_i := ((\mathbf{I}_{r_i} \otimes \mathbf{J}_i) \otimes \mathbf{I}_\varphi), \quad \mathbf{P} := \mathbf{J}_{d-1} \mathbf{C}_{d-2} \cdots \mathbf{C}_0.$$

When $d = 1$, the product of the matrices $\mathbf{C}_i$ is the identity. Then, for every $\mathbf{w} \in \mathcal{R}_q^{r_0 m_0}$,

$$\text{Tr}(\mathbf{M}\mathbf{w}) = \mathbf{P}\text{coeff}(\mathbf{w}) \bmod q.$$

Proof. For $i \in [d-1]$, the coefficient identity for the coarse lift gives

$$\text{coeff}((\mathbf{I}_{r_i} \otimes \mathbf{M}_i)\mathbf{u}) = \mathbf{C}_i \text{coeff}(\mathbf{u}) \bmod q.$$

After iterating over the coarse layers, the defining identity for the final fine lift gives

$$\text{Tr}(\mathbf{M}\mathbf{w}) = \mathbf{J}_{d-1} \mathbf{C}_{d-2} \cdots \mathbf{C}_0 \text{coeff}(\mathbf{w}) = \mathbf{P}\text{coeff}(\mathbf{w}) \bmod q.$$

$\square$

Definition 23 (Layered fine/coarse ring challenge distribution). *Let $d \in \mathbb{N}$, let q be an unramified prime for $\mathcal{R}$, and let $(r_i, m_i, n_i)_{i \in [d]}$ be positive integers satisfying*

$$r_{d-1} = 1, \quad r_i n_i = r_{i+1} m_{i+1} \quad \text{for every } i \in [d-1].$$

We write

$$\mathbf{M} \leftarrow_{\$} \chi_{\otimes \mathcal{R}; d; (r_i, n_i, m_i)_{i \in [d]}}, \quad \mathbf{M} \in \mathcal{R}_q^{n_{d-1} \times r_0 m_0}$$

to sample independently

$$\mathbf{J}_i \leftarrow_{\$} \chi^{n_i \times m_i} \quad \text{for } i \in [d-1], \quad \mathbf{J}_{d-1} \leftarrow_{\$} \chi^{n_{d-1} \times (\varphi m_{d-1})},$$

and set

$$\mathbf{M}_i := \text{Lift}_{\text{coarse}}(\mathbf{J}_i) \in \mathcal{R}_q^{n_i \times m_i} \quad \text{for } i \in [d-1], \quad \mathbf{M}_{d-1} := \text{Lift}_{\text{fine}}(\mathbf{J}_{d-1}) \in \mathcal{R}_q^{n_{d-1} \times m_{d-1}},$$

$$\mathbf{M} := (\mathbf{I}_{r_{d-1}} \otimes \mathbf{M}_{d-1}) \cdots (\mathbf{I}_{r_1} \otimes \mathbf{M}_1)(\mathbf{I}_{r_0} \otimes \mathbf{M}_0).$$

Set

$$\eta_i := \varphi r_i, \quad \hat{m}_i := m_i \quad \text{for } i \in [d-1], \quad \eta_{d-1} := 1, \quad \hat{m}_{d-1} := \varphi m_{d-1}.$$

Up to fixed coordinate permutations, the corresponding coefficient matrix $\mathbf{P}$ of Lemma 10 is distributed as the layered projection of Lemma 9 with layer dimensions $(\eta_i, n_i, \hat{m}_i)_{i \in [d]}$.

Definition 24 (Layered multilinear extension). Let $\mathbf{M} \leftarrow_{\$} \chi_{\otimes \mathcal{R}; d; (r_i, n_i, m_i)_{i \in [d]}}$ be as in Definition 23. Write

$$N_0 := r_0 m_0, \quad N_{i+1} := r_i n_i = r_{i+1} m_{i+1} \quad \text{for every } i \in [d-1], \quad N_d := n_{d-1}.$$

Note that $N_0, \dots, N_{d-1}$ count ring elements whereas N_d counts $\mathbb{Z}_q$ coordinates, since the last layer is a *fine* lift. That boundary therefore shrinks the coefficient count by $\varphi m_{d-1}/n_{d-1}$. Assume that $N_i = 2^{\nu_i}$ for every $i \in [0 : d+1]$. Thus, the i th tensor-like layer is $\mathbf{I}_{r_i} \otimes \mathbf{M}_i \in \mathcal{R}_q^{N_{i+1} \times N_i}$, and their product is the sampled matrix $\mathbf{M}$. Boundary 0 is the projection output and boundary d is the witness, so the variable sets are indexed in the opposite order to the layers: $\mathbf{x}_i$ carries ν_{d-i} variables. The *layered multilinear extension* of $\mathbf{M}$ is the polynomial with one variable set $\mathbf{x}_i \in \mathcal{R}_q^{\nu_{d-i}}$ for every layer boundary $i \in [0 : d+1]$, defined by

$$\text{LMLE}[\mathbf{M}](\mathbf{x}_0, \dots, \mathbf{x}_d) := \prod_{i \in [d]} \text{MLE}[\mathbf{I}_{r_{d-1-i}} \otimes \mathbf{M}_{d-1-i}](\mathbf{x}_i, \mathbf{x}_{i+1}).$$

It refines the MLE of the product by retaining every cross-layer index: for all $\mathbf{x}_0 \in \mathcal{R}_q^{\nu_d}$ and $\mathbf{x}_d \in \mathcal{R}_q^{\nu_0}$,

$$\text{MLE}[\mathbf{M}](\mathbf{x}_0, \mathbf{x}_d) = \sum_{\substack{\mathbf{b}_1 \in \{0,1\}^{\nu_{d-1}}, \dots, \\ \mathbf{b}_{d-1} \in \{0,1\}^{\nu_1}}} \text{LMLE}[\mathbf{M}](\mathbf{x}_0, \mathbf{b}_1, \dots, \mathbf{b}_{d-1}, \mathbf{x}_d).$$

3.2 Folding

Before describing the folding protocol, we introduce the commitment-key structure, which formally captures the idea that the matrix $\mathbf{F}$ of the principal relation is a shared commitment key. Furthermore, we define the relation collecting the solutions to the SIS problem of Definition 17 under a fixed key, which captures the alternative outcome of extraction in the folding protocol.

Definition 25 (Commitment-key structure). For every statement $(\mathbf{F}, \mathbf{y}, \mathbf{s}, t)$ of the principal relation, define

$$\text{str}(\mathbf{F}, \mathbf{y}, \mathbf{s}, t) := \mathbf{F}.$$

Definition 26 (SIS-break Relation). Let $m, n \in \mathbb{N}$, $\beta \in \mathbb{R}$. We define the relation

$$\Xi_{\mathcal{R}, q, m, n, \beta}^{\text{SIS}} := \left\{ (\mathbf{F}, \mathbf{x}) : \begin{array}{l} \mathbf{F} \in \mathcal{R}_q^{n \times m}; \mathbf{x} \in \mathcal{R}^m \\ \|\mathbf{x}\| \leq \beta; \mathbf{F}\mathbf{x} = 0; \mathbf{x} \neq 0 \end{array} \right\}$$

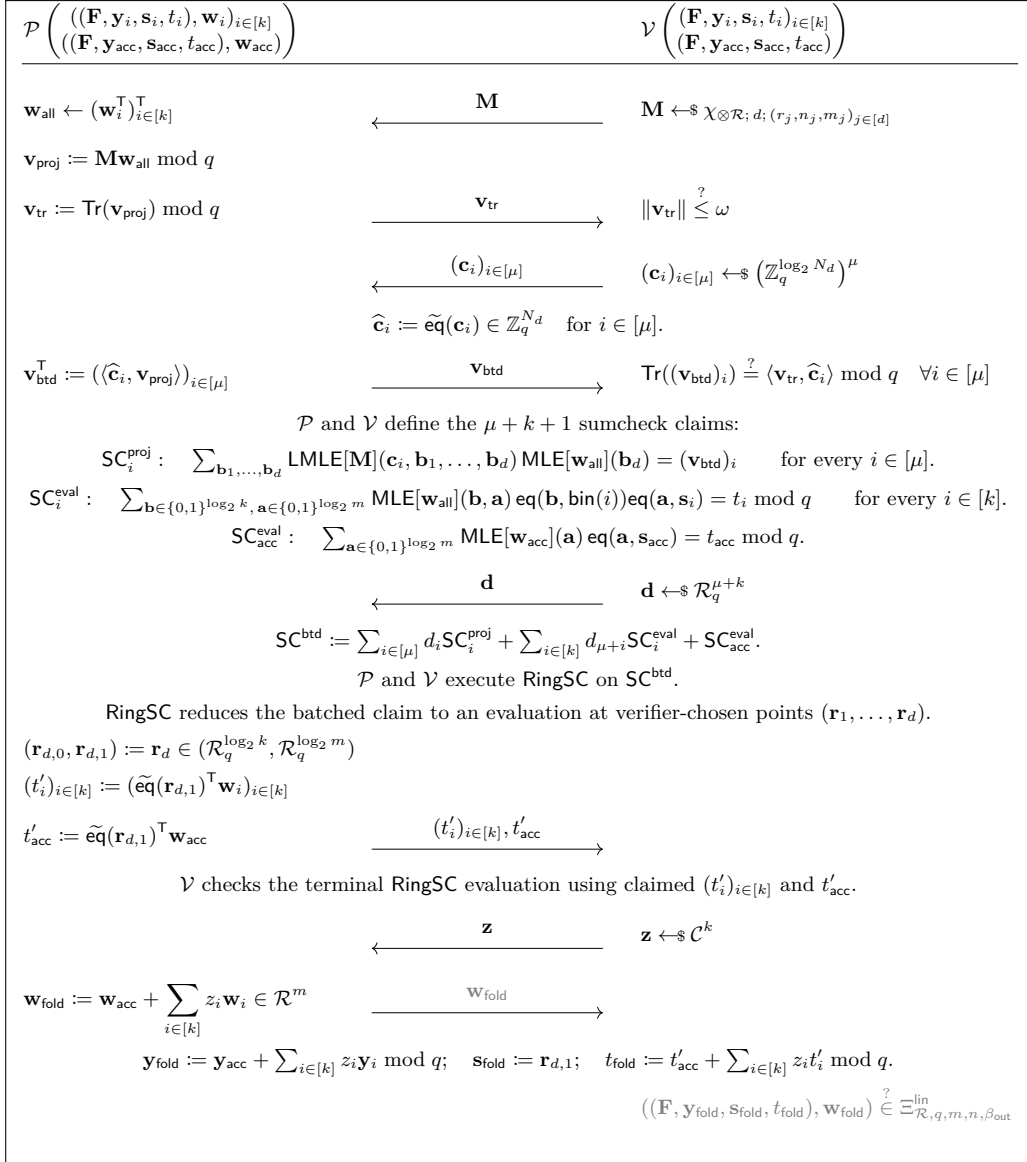


Figure 2: The folding protocol Π^{fold} .

Theorem 1 (PikkuFold folding). *Let $\mathcal{R}_q \cong (\mathbb{F}_{q^e})^{\varphi/e}$ and fix $a \mid e$ for RingSC per Lemma 8. Let k, m, μ, φ and the boundary dimensions defined below be powers of two. Let d be the number of layers in the JL projection, and let $(r_j, m_j, n_j)_{j \in [d]}$ be the layer dimensions. The layer dimensions themselves need only satisfy*

$$r_0 m_0 = km, \quad r_{d-1} = 1, \quad r_j m_j = r_{j-1} n_{j-1} \quad \text{for every } j \in [1 : d].$$

Set

$$\begin{aligned} N_0 &:= km, & N_d &:= n_{d-1}, \\ N_{j+1} &:= r_j n_j = r_{j+1} m_{j+1} & \text{for } j \in [d-1], \\ \nu_j &:= \log_2 N_j & \text{for } j \in [d+1]. \end{aligned}$$

For $j \in [d]$, let $\ell_j := \alpha_{\text{rp}}(\kappa_j)$, $u_j := \beta_{\text{rp}}(\kappa_j)$, and $b_j := b(\kappa_j)$ be the parameters of Definition 16. Assume that every n_j is chosen so that the guarantee holds for the corresponding

coefficient-projection layer with error κ_j , and define

$$\eta_j := \begin{cases} \varphi r_j, & 0 \leq j \leq d-2, \\ 1, & j = d-1, \end{cases} \quad \kappa_{\text{proj}} := \sum_{j \in [d]} \eta_j \kappa_j,$$

$$\omega := \sqrt{k} \beta_{\text{in}} \prod_{j \in [d]} u_j, \quad \beta'_{\text{in}} := \frac{\omega}{\prod_{j \in [d]} \ell_j}.$$

Assume the modular projection conditions

$$\beta'_{\text{in}} \prod_{h \in [j]} \ell_h \leq \frac{q}{b_j} \quad \text{for every } j \in [d].$$

Let $\mathcal{C} \subseteq \mathcal{R}$ be an $\epsilon_{\mathcal{C}}$ -approximate strong sampling set modulo q with expansion factor $\Gamma_{\mathcal{C}}$. Let $\beta_{\text{in}}, \beta_{\text{acc}}, \beta'_{\text{out}} > 0$ and define

$$\beta_{\text{out}} := \beta_{\text{acc}} + k\Gamma_{\mathcal{C}}\beta_{\text{in}}, \quad \beta'_{\text{acc}} := \beta'_{\text{out}} + k\Gamma_{\mathcal{C}}\beta'_{\text{in}}.$$

Define the temporary binding bounds

$$\beta_{\text{bind}} := (k+1)\beta'_{\text{out}}(2\Gamma_{\mathcal{C}})^k, \quad \rho_{\text{bind}} := (2\Gamma_{\mathcal{C}})^k, \quad \beta_{\text{sis}} := \max\{8\Gamma_{\mathcal{C}}\beta'_{\text{out}}, 2\beta_{\text{bind}}\rho_{\text{bind}}\}.$$

With the commitment-key structure maps of Definitions 25 and 26, protocol Π^{fold} in Figure 2 is a relaxed folding scheme in the sense of Definition 21. It is correct for

$$(\Xi_{\mathcal{R}, q, m, n, \beta_{\text{in}}}^{\text{lin}})^k \times \Xi_{\mathcal{R}, q, m, n, \beta_{\text{acc}}}^{\text{lin}} \longrightarrow \Xi_{\mathcal{R}, q, m, n, \beta_{\text{out}}}^{\text{lin}}$$

with correctness error κ_{proj} . It is knowledge sound for

$$\left((\Xi_{\mathcal{R}, q, m, n, \beta'_{\text{in}}}^{\text{lin}})^k \times \Xi_{\mathcal{R}, q, m, n, \beta'_{\text{acc}}}^{\text{lin}} \right) \cup \Xi_{\mathcal{R}, q, m, n, \beta_{\text{sis}}}^{\text{SIS}} \longleftarrow \Xi_{\mathcal{R}, q, m, n, \beta'_{\text{out}}}^{\text{lin}}$$

with knowledge error

$$\kappa := \frac{k}{|\mathcal{C}|} + k\epsilon_{\mathcal{C}} + \kappa_{\text{proj}} + \left(\frac{\log_2 N_d}{q} \right)^{\mu} + \frac{1}{q^e} + \frac{2 \sum_{j \in [d]} \nu_j + 1}{q^a}.$$

Proof. Correctness. Let the fresh witnesses be $(\mathbf{w}_i)_{i \in [k]}$ and let $\mathbf{w}_{\text{acc}}$ be the accumulator witness. Their concatenation satisfies

$$\|\mathbf{w}_{\text{all}}\| \leq \sqrt{k} \beta_{\text{in}}.$$

Let $\mathbf{P}$ be the integer layered matrix corresponding to $\mathbf{M}$ under Lemma 10. By the upper-bound part of Lemma 9, except with probability at most κ_{proj} ,

$$\|\mathbf{P}\text{coeff}(\mathbf{w}_{\text{all}})\| \leq \sqrt{k} \beta_{\text{in}} \prod_{j \in [d]} u_j = \omega.$$

The identity in Lemma 10 gives

$$\mathbf{v}_{\text{tr}} = \text{Tr}(\mathbf{M}\mathbf{w}_{\text{all}}) = \mathbf{P}\text{coeff}(\mathbf{w}_{\text{all}}) \bmod q,$$

and balanced reduction modulo q cannot increase its Euclidean norm. Hence the projection norm check passes outside the stated event. Every trace-batching equation holds by $\mathbb{Z}_q$ -linearity of Tr , and every projection and evaluation sumcheck claim is true. Perfect completeness of Lemma 8 implies that the terminal sumcheck check passes.

The common-point claims satisfy $t'_i = \text{MLE}[\mathbf{w}_i](\mathbf{r}_{d,1})$ and $t'_{\text{acc}} = \text{MLE}[\mathbf{w}_{\text{acc}}](\mathbf{r}_{d,1})$. Therefore

$$\text{MLE}[\mathbf{w}_{\text{fold}}](\mathbf{r}_{d,1}) = t'_{\text{acc}} + \sum_{i \in [k]} z_i t'_i = t_{\text{fold}} \bmod q.$$

Linearity also gives $\mathbf{F}\mathbf{w}_{\text{fold}} = \mathbf{y}_{\text{fold}} \bmod q$, and

$$\|\mathbf{w}_{\text{fold}}\| \leq \|\mathbf{w}_{\text{acc}}\| + \sum_{i \in [k]} \|z_i\|_{\text{op}} \|\mathbf{w}_i\| \leq \beta_{\text{acc}} + k\Gamma_C \beta_{\text{in}} = \beta_{\text{out}}.$$

The output lies in the claimed relation and has the same commitment-key structure $\mathbf{F}$.

Knowledge soundness. We first analyse a deterministic prover by fixing its coins. The general case follows by averaging over them. Let p be the probability that its output instance and witness lie in $\Xi_{\mathcal{R},q,m,n,\beta'_{\text{out}}}^{\text{lin}}$. We describe the extractor in four stages.

Folding extraction and denominator clearing. Fix the complete protocol prefix through the messages $(t'_i)_{i \in [k]}$ and t'_{acc} . Apply the coordinate-wise forking extractor of Lemma 6 to this prefix with $\ell = k$ and special-soundness parameter 2. Except for loss $k/|\mathcal{C}|$, it returns accepting transcripts for challenges

$$\mathbf{z}^{(0)}, \dots, \mathbf{z}^{(k)} \in \mathcal{C}^k$$

such that $\mathbf{z}^{(i)}$ and $\mathbf{z}^{(k)}$ differ only in coordinate i . The corresponding output witnesses $(\mathbf{u}^{(j)})_{j \in [0:k+1]}$ have norm at most β'_{out} . For $i \in [k]$, set

$$\Delta_i := z_i^{(i)} - z_i^{(k)}, \quad \mathbf{a}_i := \mathbf{u}^{(i)} - \mathbf{u}^{(k)}.$$

The equality event in each forked coordinate is charged by the $k/|\mathcal{C}|$ forking loss. After the first value in the coordinate is fixed, the second rewind samples an independent value from $\mathcal{C}$. By Definition 11, except with probability at most $k\epsilon_C$, all Δ_i are units in $\mathcal{R}_q$. They satisfy

$$\|\Delta_i\|_{\text{op}} \leq 2\Gamma_C, \quad \|\mathbf{a}_i\| \leq 2\beta'_{\text{out}}, \quad \mathbf{F}\mathbf{a}_i = \Delta_i \mathbf{y}_i \bmod q, \quad \text{MLE}[\mathbf{a}_i](\mathbf{r}_{d,1}) = \Delta_i t'_i \bmod q.$$

Let

$$\Delta := \prod_{i \in [k]} \Delta_i, \quad \Delta_{-i} := \prod_{j \in [k] \setminus \{i\}} \Delta_j, \quad \mathbf{a}_{\text{acc}} := \Delta \mathbf{u}^{(k)} - \sum_{i \in [k]} z_i^{(k)} \Delta_{-i} \mathbf{a}_i.$$

Subtracting the folded output equations and then clearing all denominators gives

$$\mathbf{F}\mathbf{a}_{\text{acc}} = \Delta \mathbf{y}_{\text{acc}} \bmod q, \quad \text{MLE}[\mathbf{a}_{\text{acc}}](\mathbf{r}_{d,1}) = \Delta t'_{\text{acc}} \bmod q.$$

Moreover,

$$\|\Delta\|_{\text{op}} \leq \rho_{\text{bind}}, \quad \|\mathbf{a}_{\text{acc}}\| \leq \beta'_{\text{out}} (2\Gamma_C)^k + k\Gamma_C (2\beta'_{\text{out}}) (2\Gamma_C)^{k-1} = \beta_{\text{bind}}.$$

Binding two independent extractions. Run the preceding extraction again with independent verifier coins until it returns another usable family, denoted by hats. If, for some $i \in [k]$,

$$\mathbf{x}_i := \mathbf{a}_i \hat{\Delta}_i - \hat{\mathbf{a}}_i \Delta_i$$

is non-zero, then

$$\mathbf{F}\mathbf{x}_i = 0 \bmod q, \quad \|\mathbf{x}_i\| \leq 8\Gamma_C \beta'_{\text{out}}.$$

The extractor outputs $(\mathbf{F}, \mathbf{x}_i)$ in the SIS-break relation. Likewise, if

$$\mathbf{x}_{\text{acc}} := \mathbf{a}_{\text{acc}} \hat{\Delta} - \hat{\mathbf{a}}_{\text{acc}} \Delta$$

is non-zero, then $\mathbf{F}\mathbf{x}_{\text{acc}} = 0 \bmod q$ and

$$\|\mathbf{x}_{\text{acc}}\| \leq 2\beta_{\text{bind}}\rho_{\text{bind}}.$$

This also gives a witness for the stated SIS-break relation.

Suppose neither case occurs. Since all denominators are units modulo q , the two extractions define the same normalised vectors

$$\mathbf{w}_i^* := \Delta_i^{-1}\mathbf{a}_i = \widehat{\Delta}_i^{-1}\widehat{\mathbf{a}}_i \bmod q, \quad \mathbf{w}_{\text{acc}}^* := \Delta^{-1}\mathbf{a}_{\text{acc}} = \widehat{\Delta}^{-1}\widehat{\mathbf{a}}_{\text{acc}} \bmod q.$$

Take balanced integral representatives for the fresh vectors and concatenate them as $\mathbf{w}_{\text{all}}^*$. The second transcript challenges are independent verifier coins with respect to the first transcript, so these normalised vectors are independent of the projection matrix, trace-batching points, subfield-batching vector, claim-batching coefficients, and sumcheck challenges of the second extraction, all of which we use below. The output relations of the second transcript family give

$$\text{MLE}[\mathbf{w}_i^*](\mathbf{r}_{d,1}) = t'_i \bmod q$$

for every $i \in [k]$, and the analogous two equalities hold for $\mathbf{w}_{\text{acc}}^*$, $\mathbf{y}_{\text{acc}}$, and t'_{acc} .

Sumcheck and projection. Evaluate the terminal RingSC claim using the fixed candidate family. The preceding common-point equalities make this terminal evaluation identical to the verifier's accepted evaluation. Let $h := \mu + k + 1$, and denote the initial ring claims by

$$\sum_{\mathbf{b} \in \{0,1\}^{\ell_u}} g_u(\mathbf{b}) = t_u \quad \text{for } u \in [h],$$

where $u = h - 1$ is the accumulator claim and $\ell := \max_{u \in [h]} \ell_u = \sum_{j \in [d]} \nu_j$. For every $u \in [h]$, set

$$\bar{g}_u(\mathbf{x}_0, \dots, \mathbf{x}_{\ell-1}) := \left(\prod_{j \in [\ell-\ell_u]} (1 - \mathbf{x}_j) \right) g_u(\mathbf{x}_{\ell-\ell_u}, \dots, \mathbf{x}_{\ell-1}).$$

For $\mathbf{d} \leftarrow \mathcal{R}_q^{h-1}$, the claim supplied to RingSC is

$$\sum_{\mathbf{b} \in \{0,1\}^{\ell}} H(\mathbf{b}) = S, \quad H := \sum_{u \in [h-1]} d_u \bar{g}_u + \bar{g}_{h-1}, \quad S := \sum_{u \in [h-1]} d_u t_u + t_{h-1}.$$

The padding preserves every sum because $\sum_{b \in \{0,1\}} (1 - b) = 1$. If at least one initial ring claim is false, then the batched claim is false except with probability q^{-e} . Indeed, fix an NTT slot containing a non-zero initial discrepancy. Conditional on the remaining coefficients, either one uniform slot of one d_u has exactly one cancelling value or only the fixed-coefficient accumulator discrepancy remains and cancellation is impossible. By Lemma 8 and because the individual degrees of the sumcheck polynomial are at most 2, the false batched claim is accepted with probability at most

$$\frac{2 \sum_{j \in [d]} \nu_j + 1}{q^a}.$$

Consequently, except with probability

$$\frac{1}{q^e} + \frac{2 \sum_{j \in [d]} \nu_j + 1}{q^a},$$

all $\mu + k + 1$ initial ring claims are true. In particular,

$$\text{MLE}[\mathbf{w}_i^*](\mathbf{s}_i) = t_i \bmod q \quad \text{for every } i \in [k], \quad \text{MLE}[\mathbf{w}_{\text{acc}}^*](\mathbf{s}_{\text{acc}}) = t_{\text{acc}} \bmod q,$$

and, for every $i \in [\mu]$,

$$(\mathbf{v}_{\text{btd}})_i = \hat{\mathbf{c}}_i^\top \mathbf{M} \mathbf{w}_{\text{all}}^* \bmod q.$$

Taking traces in the last equality gives

$$\text{Tr}((\mathbf{v}_{\text{btd}})_i) = \hat{\mathbf{c}}_i^\top \text{Tr}(\mathbf{M} \mathbf{w}_{\text{all}}^*) \bmod q.$$

The accepted trace check gives

$$\text{Tr}((\mathbf{v}_{\text{btd}})_i) = \hat{\mathbf{c}}_i^\top \mathbf{v}_{\text{tr}} \bmod q,$$

and hence

$$\hat{\mathbf{c}}_i^\top (\text{Tr}(\mathbf{M} \mathbf{w}_{\text{all}}^*) - \mathbf{v}_{\text{tr}}) = 0 \bmod q \quad \text{for every } i \in [\mu].$$

If the vector in parentheses is non-zero, its multilinear extension is a non-zero multilinear polynomial in $\log_2 N_d$ variables. The points $(\mathbf{c}_i)_{i \in [\mu]}$ are independent and uniform over $\mathbb{Z}_q^{\log_2 N_d}$, so Lemma 7 bounds the probability that all equalities hold by

$$\left(\frac{\log_2 N_d}{q} \right)^\mu.$$

Outside this event, Lemma 10 gives

$$\mathbf{P} \text{coeff}(\mathbf{w}_{\text{all}}^*) = \text{Tr}(\mathbf{M} \mathbf{w}_{\text{all}}^*) = \mathbf{v}_{\text{tr}} \bmod q.$$

The vector $\mathbf{w}_{\text{all}}^*$ is fixed before $\mathbf{P}$ is sampled, and the verifier accepted only if $\|\mathbf{v}_{\text{tr}}\| \leq \omega$. The modular lower-bound part of Lemma 9, applied with threshold β'_{in} , now gives

$$\|\mathbf{w}_{\text{all}}^*\| \leq \beta'_{\text{in}}$$

except with probability at most κ_{proj} . Hence every fresh candidate has norm at most β'_{in} and is a valid witness for its input relation.

Additive accumulator reconstruction. Return to the baseline transcript and define the integral vector

$$\tilde{\mathbf{w}}_{\text{acc}} := \mathbf{u}^{(k)} - \sum_{i \in [k]} z_i^{(k)} \mathbf{w}_i^*.$$

Modulo q , this vector equals $\mathbf{w}_{\text{acc}}^*$. It therefore satisfies the accumulator's linear and original evaluation claims established above. Its norm satisfies

$$\|\tilde{\mathbf{w}}_{\text{acc}}\| \leq \beta'_{\text{out}} + k\Gamma_C \beta'_{\text{in}} = \beta'_{\text{acc}}.$$

The extractor outputs the k fresh witnesses and $\tilde{\mathbf{w}}_{\text{acc}}$.

The union bound over coordinate-wise extraction, non-unit differences, ring sumcheck, trace batching, and the modular projection gives the stated knowledge error. Thus the extractor returns a witness for the input relation or the same-key SIS-break relation with probability at least $p - \kappa$.

Expected runtime of the extractor. Let p_0 be the probability that one invocation of the coordinate-wise extractor returns a usable family with invertible differences. One invocation is made unconditionally. If $p_0 = 0$, the extractor stops after this invocation. Otherwise, conditional on success, independent invocations are made until a second usable family is obtained. The expected number of family-extractor invocations is

$$(1 - p_0) + p_0 \left(1 + \frac{1}{p_0} \right) = 2.$$

By Lemma 6, instantiated with $\ell = k$ and special-soundness parameter 2, one family extraction uses at most $k + 1$ expected invocations of the prover. All remaining operations are polynomial in the statement and witness dimensions. Hence the extractor runs in expected polynomial time. $\square$

Remark 3 (Improved extraction strategy). Subsequent to the publication of **PikkuFold**, a refined extraction strategy for Cyclo [GLLO26] was published [WF26], which allows us to replace the SIS-break bound $\beta_{\text{sis}} := \max \{8\Gamma_C \beta'_{\text{out}}, 2\beta_{\text{bind}} \rho_{\text{bind}}\}$ with the smaller bound $6\Gamma_C \beta'_{\text{out}}$. The refinement relies on rewinding the prover with fixed *folding challenges* $\mathbf{z}$. Because it requires a more involved statistical argument, we present this extension in Appendix A and keep the original analysis in Theorem 1 for clarity.

3.3 Efficiency

For the following statement, define

$$L_{\text{sc}} := \sum_{j \in [d]} \nu_j, \quad S_{\text{lay}} := \sum_{j \in [d]} r_j n_j m_j, \quad S_{\text{JL}} := \sum_{j \in [d-1]} n_j m_j + \varphi n_{d-1} m_{d-1}.$$

Thus, S_{lay} is the number of non-zero ring entries in the expanded block-diagonal layers, whereas S_{JL} is the number of ternary entries in their unexpanded JL descriptions.

Lemma 11 (Efficiency of **PikkuFold**). *The protocol of Figure 2 has the following costs, for arbitrary layer dimensions satisfying the compatibility constraints of Theorem 1.*

1. The prover uses

$$O(S_{\text{lay}} + (k+1)m + \mu N_d + km + kn)$$

operations over $\mathcal{R}_q$, in addition to

$$O(S_{\text{JL}} + \varphi N_d)$$

coefficient operations for the fine lift and the trace, and

$$O\left(L_{\text{sc}} \frac{\varphi}{a}\right)$$

operations over $\mathbb{F}_{q^a}$ for the subfield batching of the sumcheck messages.

2. The verifier uses

$$O(S_{\text{JL}} + \mu N_d + k \log_2 m + kn)$$

operations over $\mathcal{R}_q$,

$$O(\mu N_d)$$

operations over $\mathbb{Z}_q$, and

$$O\left(L_{\text{sc}} + \frac{\varphi}{a}\right)$$

operations over $\mathbb{F}_{q^a}$.

3. Excluding the output witness, which is the prover's local output in the RoK interface, prover-to-verifier communication is

$$N_d \text{ elements of } \mathbb{Z}_q, \quad \mu + k + 1 \text{ elements of } \mathcal{R}_q, \quad 2L_{\text{sc}} \text{ elements of } \mathbb{F}_{q^a}.$$

Equivalently, writing $b_q := \lceil \log_2 q \rceil$, prover-to-verifier communication is

$$(N_d + \varphi(\mu + k + 1) + 2aL_{\text{sc}})b_q$$

bits.

Proof. The prover applies the layers successively without materialising their zero blocks. The j th application processes $r_j n_j m_j$ ring entries. For the projection sumchecks, first form the linear combination of the μ expanded output-point vectors in $O(\mu N_d)$ operations. The resulting sumcheck polynomial is a chain

$$L_0(\mathbf{x}_1) \prod_{j \in [1:d]} \text{MLE}[\mathbf{I}_{r_{d-1-j}} \otimes \mathbf{M}_{d-1-j}](\mathbf{x}_j, \mathbf{x}_{j+1}) \text{MLE}[\mathbf{w}_{\text{all}}](\mathbf{x}_d),$$

with the evident omission of the displayed product when $d = 1$. Standard table folding processes one boundary at a time. Once the verifier fixes that boundary, its table is contracted into the adjacent layer and discarded. Across all rounds, every non-zero layer entry and every witness entry is processed only a constant number of times for each of the two transmitted univariate coefficients. The k fresh evaluation claims and the accumulator claim add tables of total length $(k+1)m$. This gives the first term in the prover bound. Computing the folded witness and output statement contributes $km + kn$ operations. Applying the dense subfield batching map to the $2L_{\text{sc}}$ sumcheck elements gives the stated $\mathbb{F}_{q^a}$ cost.

At the terminal point, the verifier uses

$$\text{MLE}[\mathbf{I}_{r_j} \otimes \mathbf{M}_j](\mathbf{x}_0, \mathbf{x}_1, (\mathbf{y}_0, \mathbf{y}_1)) = \text{eq}(\mathbf{x}_0, \mathbf{y}_0) \text{MLE}[\mathbf{M}_j](\mathbf{x}_1, \mathbf{y}_1).$$

For the μ projection claims, the verifier constructs the vectors $\hat{\mathbf{c}}_i$ and their tensor factors in $O(\mu N_d)$ operations over $\mathbb{Z}_q$. It then forms the ring-valued batched output weight $\sum_{i \in [\mu]} d_i \hat{\mathbf{c}}_i$ and scans each unexpanded matrix description once. Thus, the verifier evaluates the layered matrix only from the S_{JL} -entry descriptions and never materialises either the expanded block-diagonal layers or their product. Forming the batched output weight and checking the μ trace equations costs $O(\mu N_d)$ operations over $\mathcal{R}_q$. The terminal input-evaluation claims cost $O(k \log_2 m)$ operations over $\mathcal{R}_q$, and computing the folded output statement costs $O(kn)$ operations over $\mathcal{R}_q$. The round checks and the terminal subfield batching have the stated field-operation counts.

It remains to account for the terminal ring element in Definition 15. At the terminal point,

$$\text{MLE}[\mathbf{w}_{\text{all}}](\mathbf{r}_{d,0}, \mathbf{r}_{d,1}) = \sum_{i \in [k]} \tilde{\text{eq}}(\mathbf{r}_{d,0})_i t'_i, \quad \text{MLE}[\mathbf{w}_{\text{acc}}](\mathbf{r}_{d,1}) = t'_{\text{acc}}.$$

All remaining factors of the terminal batched polynomial are public. Hence the verifier reconstructs the terminal ring element from $(t'_i)_{i \in [k]}$ and t'_{acc} , and the generic terminal ring-element message of RingSC is omitted. Therefore the prover sends only the μ elements of $\mathbf{v}_{\text{btd}}$ and these $k+1$ terminal values over $\mathcal{R}_q$, giving the stated communication count. $\square$

Corollary 1 (Constant-factor layer shrinkage). *Suppose that every boundary shrinks by the same factor $\varsigma > 1$, so that $N_{j+1} = N_j/\varsigma$, and that $n_j = n_0$ for every $j \in [d]$. Then $m_j = \varsigma n_0$ for every $j \in [d]$, $N_d = n_0$, and*

$$d = \log_{\varsigma} \left(\frac{N_0}{n_0} \right), \quad L_{\text{sc}} = d \log_2 N_0 - \frac{d(d-1)}{2} \log_2 \varsigma.$$

$$S_{\text{lay}} = n_0 \sum_{j \in [d]} N_j = n_0 N_0 \frac{1 - \varsigma^{-d}}{1 - \varsigma^{-1}} \leq n_0 N_0 \frac{\varsigma}{\varsigma - 1}, \quad S_{\text{JL}} = \varsigma n_0^2 (d - 1 + \varphi).$$

Consequently, the prover uses

$$O \left(n_0 N_0 \frac{\varsigma}{\varsigma - 1} + (k+1)m + \mu n_0 + km + kn \right)$$

operations over $\mathcal{R}_q$, together with

$$O(\varsigma n_0^2(d-1+\varphi) + \varphi n_0)$$

coefficient operations and $O(L_{\text{sc}}\varphi/a)$ operations over $\mathbb{F}_{q^a}$. The verifier uses

$$O(\varsigma n_0^2(d-1+\varphi) + \mu n_0 + k \log_2 m + kn)$$

operations over $\mathcal{R}_q$, $O(\mu n_0)$ operations over $\mathbb{Z}_q$, and $O(L_{\text{sc}} + \varphi/a)$ operations over $\mathbb{F}_{q^a}$.

Proof. The identities for d and L_{sc} follow from $N_j = N_0/\varsigma^j$. Moreover, $N_{j+1}/N_j = n_j/m_j = 1/\varsigma$, so $m_j = \varsigma n_0$, and $r_{d-1} = 1$ gives $N_d = n_{d-1} = n_0$. Since $r_j m_j = N_j$, we have $r_j n_j m_j = N_j n_0$, and summing the resulting geometric sequence gives the expression for S_{lay} . Substitution into the definition of S_{JL} gives $S_{\text{JL}} = \varsigma n_0^2(d-1+\varphi)$. The runtime bounds follow from Lemma 11. $\square$

4 A Ternary Johnson–Lindenstrauss Projection

In recent years, the Johnson–Lindenstrauss lemma has been used extensively in lattice-based succinct arguments, yet its concrete parameters there have been justified only heuristically. The reason for its popularity is that these projections can be used to argue about the norm of a vector without revealing it.

The first use of this Johnson–Lindenstrauss norm certificate for proving shortness appeared in [GHL22], which also gave a concrete heuristic analysis of its parameters. Later, the projection was equipped with a modular counterpart in [BS23], which permits norm arguments when the projection is computed modulo q . Since then, the ternary Johnson–Lindenstrauss projection has been used in several works, including [LNP22, BS23, GHL22, NS24, LSS24, KLNO25, KLOT25, KLN+26, BBL+26b, Osa26, BBL+26a]. The construction of [LSS24] has also been submitted to the NIST Threshold Call [LNO+26].

The intuition that the ternary Johnson–Lindenstrauss projection approximately preserves norms is well known and holds asymptotically. Its concrete parameters at cryptographic security levels, however, are difficult to verify directly. The broad use of this projection in cryptography motivates a rigorous analysis of its concrete parameters, which is the main contribution of this section.

In this section, we provide such an analysis for the parameters relevant to cryptography, namely $\lambda \in \{64, 96, 128, 192, 256\}$ and projections with 2λ rows. Several steps require finite calculations at these concrete parameters, so we refrain from providing exact numerical constants for arbitrary choices of λ and row count. An accompanying Sage notebook can be used to derive bounds for other choices of λ and the number of rows.¹

The ternary distribution makes the projection efficient to evaluate. Our analysis establishes four estimates. For a row $\mathbf{u}_j$ of $\mathbf{J}$, it bounds the individual inner product, the upper and lower tails of the projected norm, and the lower tail after reduction modulo q . Outside the respective failure events, these estimates have the form

$$|\langle \mathbf{w}, \mathbf{u}_j \rangle| \leq \gamma_1 \|\mathbf{w}\|, \quad \alpha \|\mathbf{w}\|^2 < \|\mathbf{J}\mathbf{w}\|^2 \leq \beta \|\mathbf{w}\|^2,$$

and

$$\|\mathbf{J}\mathbf{w} \bmod q\| > \sqrt{\alpha} \theta \quad \text{when} \quad \|\mathbf{w}\| \geq \theta.$$

The single-row estimate is needed separately because it controls wrap-around in the modular argument. The single-row estimate follows from a comparison with Rademacher sums. A ternary entry has the decomposition

$$u = \frac{\varepsilon + \varepsilon'}{2}$$

¹The notebook is https://github.com/osdnk/pikku/blob/main/ternary_jl_parameters.ipynb.

for independent uniform signs $\varepsilon, \varepsilon' \in \{-1, 1\}$. Hence $\langle \mathbf{w}, \mathbf{u} \rangle$ is a Rademacher sum after rescaling, and the comparison theorem of [BD15] gives the bound γ_1 . The upper tail follows from a direct moment comparison. Every even moment of a ternary entry is bounded by the corresponding moment of a Gaussian with variance $1/2$, so expanding $\|\mathbf{J}\mathbf{w}\|^{2p}$ term by term reduces the calculation to moments of $(1/2)\chi_{2\lambda}^2$.

The lower tail is the main difficulty. A Gaussian substitution is not valid uniformly over $\mathbf{w}$. For a single-coordinate unit vector, $\|\mathbf{J}\mathbf{w}\|^2$ is binomial, and its deep lower tail is heavier than that of $(1/2)\chi_{2\lambda}^2$. By homogeneity, we normalise $\|\mathbf{w}\| = 1$ and study the lower-tail transform

$$\mathcal{L}_{\mathbf{w}}(s) := \mathbb{E} \left[e^{-s \langle \mathbf{w}, \mathbf{u} \rangle^2} \right].$$

Fourier inversion rewrites it as

$$\mathcal{L}_{\mathbf{w}}(s) = \mathbb{E}_{\xi \sim N(0, 2s)} \left[\prod_{j=1}^m \cos^2 \left(\frac{w_j \xi}{2} \right) \right].$$

This identity exposes the dependence on the coordinates of $\mathbf{w}$. We call a coordinate large if $|w_j| > 0.45$ and small otherwise. On the interval $|y| \leq \pi/2$, the inequality $\cos^2 y \leq e^{-y^2}$ controls the small coordinates; the contribution outside this interval is bounded separately. The large coordinates are retained exactly. There are at most four large coordinates, since five such coordinates would have squared norm greater than one. The cases with zero or one large coordinate are handled analytically, and the cases with two, three, or four are reduced to a finite calculation.

Our bound on the transform uses the envelope

$$\text{env}(s) := \max \left\{ \frac{1 + e^{-s}}{2}, (1 + s)^{-1/2} \right\}.$$

The two terms meet at $s_{\times}$, as shown in Figure 5. Bounding the transform at this point, including the Gaussian tail from the interval restriction, gives the lower-tail constant α via the Chernoff bound. For the reduction modulo q , we follow the three-regime proof of [BS23, Appendix A] and instantiate it with the real and single-row bounds above.

The remainder of the section makes these estimates precise. The single-row and upper-tail bounds in Sections 4.1 and 4.2 are analytic. The lower-tail proof in Section 4.3 combines analytic reductions with the finite calculation in Proposition 1. The modular proof in Section 4.4 follows the reduction of [BS23, Appendix A].

Theorem 2. *Let χ be the distribution on $\{-1, 0, 1\}$ given by*

$$\Pr[\chi = 0] = \frac{1}{2}, \quad \Pr[\chi = 1] = \Pr[\chi = -1] = \frac{1}{4}.$$

For each of the five security levels in Table 1, we certify the constants $(\gamma_1, \alpha, \beta, b)$ given in the corresponding row. For every $m \in \mathbb{N}$ and every non-zero $\mathbf{w} \in \mathbb{R}^m$, sample $\mathbf{J} \in \{-1, 0, 1\}^{2\lambda \times m}$ with independent χ entries and denote its rows by $(\mathbf{u}_j)_{j \in [2\lambda]}$. Then, for every $j \in [2\lambda]$,

$$\Pr[|\langle \mathbf{w}, \mathbf{u}_j \rangle| > \gamma_1 \|\mathbf{w}\|] \leq \frac{2^{-(\lambda+1)}}{2\lambda}, \quad \Pr[\|\mathbf{J}\mathbf{w}\|^2 > \beta \|\mathbf{w}\|^2] \leq 2^{-(\lambda+1)},$$

and

$$\Pr[\|\mathbf{J}\mathbf{w}\|^2 \leq \alpha \|\mathbf{w}\|^2] \leq 2^{-(\lambda+1)}.$$

Consequently,

$$\Pr \left[\frac{\|\mathbf{J}\mathbf{w}\|}{\|\mathbf{w}\|} \notin [\sqrt{\alpha}, \sqrt{\beta}] \right] \leq 2^{-\lambda}.$$

Moreover, for every integer $q \geq 2$, every $0 < \theta \leq q/b$, and every $\mathbf{w} \in [\pm q/2]^m$ satisfying $\|\mathbf{w}\| \geq \theta$,

$$\Pr [\|\mathbf{J}\mathbf{w} \bmod q\| \leq \sqrt{\alpha}\theta] \leq 2^{-\lambda}.$$

Table 1: Concrete ternary JL constants for 2λ rows.

λ	2λ	γ_1	α	β	b
64	128	6.97	13.53	171.8	73
96	192	8.43	20.45	257.5	100
128	256	9.66	27.37	343.2	126
192	384	11.75	41.21	514.6	176
256	512	13.51	55.05	686.0	224

The constants in Table 1 are rounded conservatively: γ_1 and β upwards, α downwards, and b to the next integer.

Remark 4 (Choice of Row Count). We use 2λ rows to retain useful concrete norm bounds at failure probability $2^{-\lambda}$. For the spike $\mathbf{w} = \mathbf{e}_1$, the event $\mathbf{J}\mathbf{w} = 0$ already has probability 2^{-N} with N rows, so $N = \lambda$ consumes the entire failure budget. More quantitatively, our transform bound gives a positive α only once $N \ln(\Psi^{-1}) > (\lambda + 1) \ln 2$. Asymptotically, this requires $N > 1.1623\lambda$, while $N = 2\lambda$ yields the constants in Table 1.

Remark 5 (Asymptotics). The lower-tail constant is affine in λ :

$$\alpha(\lambda) = \frac{2\lambda \ln(\Psi^{-1}) - (\lambda + 1) \ln 2}{s_{\times}} \approx 0.216\lambda - 0.300, \quad \Psi = 0.5508,$$

where $\Psi = 0.5508$ is from Proposition 2, and the displayed coefficients are rounded. As $\lambda \rightarrow \infty$, $\beta(\lambda) \sim c\lambda$, where $c \approx 2.68$ is the root of $c - \ln c = 1 + \ln 2$, and $\gamma_1(\lambda) \sim \sqrt{\lambda} \ln 2$. For the modular construction below, extend the parameters to arbitrary λ by taking T_λ maximal subject to the binomial inequality in Equation (4), fixing any $r > \sqrt{1 + 16(0.52)^2}$, and setting $D_\lambda = \gamma_1(\lambda) + 1$. Binomial large deviations give $T_\lambda = \Theta(\lambda)$, while $D_\lambda = \Theta(\sqrt{\lambda})$ and $\ell_\lambda = \sqrt{\alpha(\lambda)} = \Theta(\sqrt{\lambda})$. The three lower bounds on $b(\lambda)$ in Equation (4) are respectively $\Theta(\lambda)$, $\Theta(\sqrt{\lambda})$, and $\Theta(\lambda)$. Thus $b(\lambda) = \Theta(\lambda)$ and the admissible modular threshold is $q/b(\lambda) = \Theta(q/\lambda)$.

4.1 Single-Row Bound

The single-row estimate bounds the largest coordinate of $\mathbf{J}\mathbf{w}$ and controls wrap-around in the modular proof. The decomposition below reduces it to a dimension-free tail bound for a Rademacher sum.

Fact 1. Each entry of a row $\mathbf{u}$ distributed as in Theorem 2 decomposes as $u_i = (\varepsilon_i + \varepsilon'_i)/2$, where $\varepsilon_i, \varepsilon'_i$ are independent and uniform on $\{-1, 1\}$. Hence $\mathbb{E}[u_i^{2p}] = 1/2$ for every $p \geq 1$, and

$$\langle \mathbf{w}, \mathbf{u} \rangle = \frac{1}{2} \sum_i w_i (\varepsilon_i + \varepsilon'_i)$$

is a Rademacher sum with coefficient norm $\|\mathbf{w}\|/\sqrt{2}$.

Lemma 12 ([BD15]). Let $S = \sum_i a_i \varepsilon_i$, where the ε_i are independent and uniform on $\{-1, 1\}$ and $\sum_i a_i^2 = 1$, and let $Z \sim N(0, 1)$. Then

$$\Pr[S \geq x] \leq \kappa_{\text{BD}} \Pr[Z \geq x]$$

for every x , where

$$\kappa_{\text{BD}} := (4 \Pr[Z \geq \sqrt{2}])^{-1} = 3.178 \dots,$$

and this constant is optimal.

Proof of the single-row bound. By homogeneity, suppose that $\|\mathbf{w}\| = 1$. By Fact 1, $S := \sqrt{2}\langle \mathbf{w}, \mathbf{u}_j \rangle$ is a Rademacher sum whose coefficient vector has Euclidean norm one, and the event is $|S| > \gamma_1 \sqrt{2}$. Applying Lemma 12 to S and $-S$ gives

$$\Pr[|S| > \gamma_1 \sqrt{2}] \leq 2\kappa_{\text{BD}} \Pr[Z \geq \gamma_1 \sqrt{2}] = \kappa_{\text{BD}} \operatorname{erfc}(\gamma_1) \leq \frac{2^{-(\lambda+1)}}{2\lambda}.$$

Here $\operatorname{erfc}(x) = 2\pi^{-1/2} \int_x^\infty e^{-t^2} dt$, and the last inequality follows from

$$\gamma_1(\lambda) = \operatorname{erfc}^{-1}\left(\frac{2^{-(\lambda+1)}}{2\lambda\kappa_{\text{BD}}}\right).$$

□

4.2 Upper Tail

The upper tail follows from a moment comparison. The even moments of the ternary distribution are bounded by those of a Gaussian with variance $1/2$, and this domination is preserved when the contributions of the independent rows are summed.

Lemma 13 (Moment domination). *Let $\mathbf{J}$ be distributed as in Theorem 2, and let $\mathbf{G} \in \mathbb{R}^{2\lambda \times m}$ have independent $N(0, 1/2)$ entries. For every $\mathbf{w} \in \mathbb{R}^m$ and every integer $p \geq 1$,*

$$\mathbb{E}[(\|\mathbf{J}\mathbf{w}\|^2)^p] \leq \mathbb{E}[(\|\mathbf{G}\mathbf{w}\|^2)^p].$$

Proof. For a single entry, $\mathbb{E}[u^{2j}] = 1/2 \leq (2j-1)!! 2^{-j} = \mathbb{E}[g^{2j}]$ for $g \sim N(0, 1/2)$, and all odd moments vanish for both. In

$$\mathbb{E}[\langle \mathbf{w}, \mathbf{u} \rangle^{2p}] = \sum_{a_1 + \dots + a_m = 2p} \binom{2p}{a_1, \dots, a_m} \prod_i w_i^{a_i} \mathbb{E}[u^{a_i}],$$

only all-even profiles survive. On each such profile, the multinomial coefficient and $\prod_i w_i^{a_i} \geq 0$ agree for $\mathbf{u}$ and the corresponding Gaussian row $\mathbf{g}$, while the product of moments is no larger for $\mathbf{u}$. Thus

$$\mathbb{E}[\langle \mathbf{w}, \mathbf{u} \rangle^{2p}] \leq \mathbb{E}[\langle \mathbf{w}, \mathbf{g} \rangle^{2p}].$$

Expanding the p th power of the sum over the 2λ independent rows lifts the inequality term by term. □

Proof of the upper-tail bound. By homogeneity, suppose that $\|\mathbf{w}\| = 1$. Then $\|\mathbf{G}\mathbf{w}\|^2 \sim \frac{1}{2}\chi_{2\lambda}^2$. Markov's inequality and Lemma 13 give

$$\Pr[\|\mathbf{J}\mathbf{w}\|^2 > \beta] \leq \frac{\mathbb{E}[(\|\mathbf{J}\mathbf{w}\|^2)^p]}{\beta^p} \leq \frac{\mathbb{E}[(\frac{1}{2}\chi_{2\lambda}^2)^p]}{\beta^p} = \frac{\Gamma(\lambda + p)}{\Gamma(\lambda)\beta^p}.$$

Minimising over integers $p \geq 1$ at the value $\beta = \beta(\lambda)$ in Table 1 gives the claimed bound $2^{-(\lambda+1)}$. □

4.3 Lower Tail

By homogeneity, it suffices to prove the lower-tail bound for $\|\mathbf{w}\| = 1$. We first associate a Laplace transform with each such vector.

Definition 27 (Lower-Tail Transform). Let $\mathbf{w}$ be a unit vector and let $\mathbf{u}$ have the row distribution of $\mathbf{J}$ in Theorem 2. For $s \geq 0$, define

$$\mathcal{L}_{\mathbf{w}}(s) := \mathbb{E}[e^{-s\langle \mathbf{w}, \mathbf{u} \rangle^2}].$$

For independent rows and every $s > 0$, Chernoff's inequality gives

$$\Pr[\|\mathbf{J}\mathbf{w}\|^2 \leq \alpha] \leq e^{\alpha s} \mathcal{L}_{\mathbf{w}}(s)^{2\lambda}. \quad (2)$$

The uniform bound proved below is

$$\sup_{\|\mathbf{w}\|=1} \mathcal{L}_{\mathbf{w}}(s_{\times}) \leq 0.5508.$$

The obstruction to a Gaussian substitution is the single-coordinate vector: its squared image norm is $\text{Bin}(2\lambda, 1/2)$, not $(1/2)\chi_{2\lambda}^2$, and in the relevant deep-tail regime its left tail is heavier, as shown in Figure 3 for $\lambda = 128$.

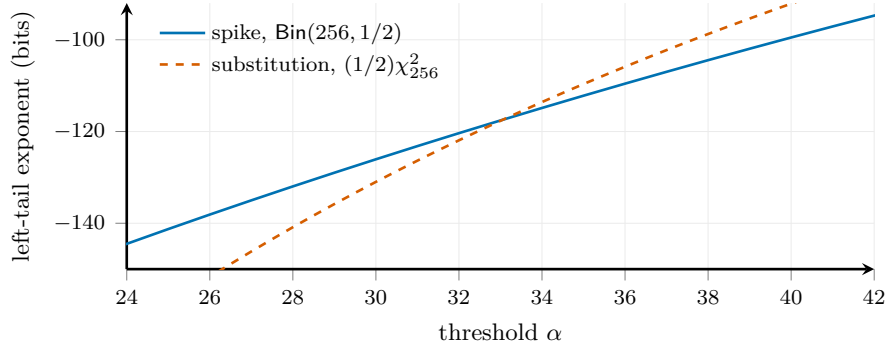


Figure 3: Real and Gaussian lower tails at $2\lambda = 256$.

Lemma 14. $\cos^2 y \leq e^{-y^2}$ for $y \in [0, \pi/2]$, with equality only at $y = 0$.

Proof. For $y \in [0, \pi/2)$, set $f(y) = \log \sec y - y^2/2$. Then $f(0) = 0$ and $f'(y) = \tan y - y > 0$, so $f \geq 0$. At $y = \pi/2$, the left side is zero. $\square$

Beyond $\pi/2$, the bound fails, as shown in Figure 4. This is why the contribution of a large coordinate is kept explicit below.

Lemma 15 (Fourier identity). For a unit vector $\mathbf{w}$ and $\xi \sim N(0, 2s)$,

$$\mathcal{L}_{\mathbf{w}}(s) = \mathbb{E}_{\xi} \left[\prod_{j=1}^m \cos^2 \left(\frac{w_j \xi}{2} \right) \right].$$

Proof. The identity $e^{-sx^2} = \mathbb{E}_{\xi \sim N(0, 2s)}[e^{i\xi x}]$, applied to $x = \langle \mathbf{w}, \mathbf{u} \rangle$, permits exchanging the expectations. The result follows from

$$\mathbb{E}[e^{i\theta u}] = \frac{1 + \cos \theta}{2} = \cos^2(\theta/2)$$

for each coordinate. $\square$

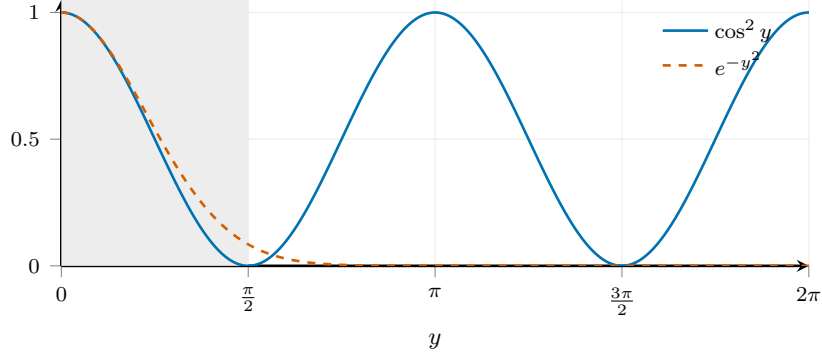


Figure 4: The cosine and Gaussian bounds.

Definition 28 (Envelope). Define

$$\text{env}(s) := \max \left\{ \frac{1 + e^{-s}}{2}, (1 + s)^{-1/2} \right\}.$$

Its branches meet at the unique $s_{\times} > 0$ satisfying

$$(1 + s_{\times})(1 + e^{-s_{\times}})^2 = 4.$$

Numerically, $s_{\times} \approx 2.31057$ and $\text{env}(s_{\times}) \approx 0.549602$.

The branches are $\mathcal{L}_{\mathbf{e}_1}(s) = (1 + e^{-s})/2$ and the limit $(1 + s)^{-1/2}$ along equal-coordinate vectors as the dimension grows.

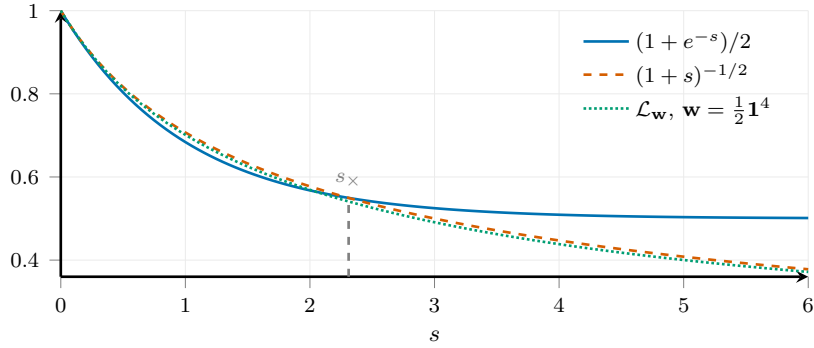


Figure 5: The transform envelope.

Lemma 16 (Peeling). Fix $\mu_0 \in (0, 1]$, a unit vector $\mathbf{w}$, and

$$L := \{j : |w_j| > \mu_0\}, \quad K := \lfloor \mu_0^{-2} \rfloor, \quad \rho^2 := \sum_{j \notin L} w_j^2.$$

Then $|L| \leq K$, and, for $s > 0$,

$$\mathcal{L}_{\mathbf{w}}(s) \leq \Phi_L(s) + \Pr_{\xi \sim N(0, 2s)}[|\xi| > \pi/\mu_0],$$

where

$$\Phi_L(s) := \mathbb{E}_{\xi \sim N(0, 2s)} \left[\left(\prod_{j \in L} \cos^2 \frac{w_j \xi}{2} \right) e^{-\rho^2 \xi^2 / 4} \right].$$

Proof. Split the expectation of Lemma 15 at $|\xi| \leq \pi/\mu_0$. On this event, $|w_j\xi/2| \leq \pi/2$ for $j \notin L$, so Lemma 14 gives

$$\prod_{j \notin L} \cos^2 \frac{w_j \xi}{2} \leq e^{-\rho^2 \xi^2 / 4}.$$

Extending the non-negative integrand to $\mathbb{R}$ gives $\Phi_L(s)$. On the complementary event, bound the product by one. $\square$

Evaluating the Gaussian expectation gives

$$\Phi_L(s) = A^{-1/2} \sum_{\sigma \in \{-1, 0, 1\}^{|L|}} \left(\prod_{j \in L} \Pr[\chi = \sigma_j] \right) \exp \left(-\frac{s}{A} \left(\sum_{j \in L} \sigma_j w_j \right)^2 \right), \quad (3)$$

$$A := 1 + \rho^2 s.$$

Thus $\Phi_\emptyset(s) = (1+s)^{-1/2} \leq \text{env}(s)$, and Φ_L depends on L only through the magnitudes $|w_j|$.

Lemma 17 (One dominant coordinate). *If $|L| = 1$ with magnitude $a \in (\mu_0, 1]$, then $\Phi_L(s) \leq \text{env}(s)$ for every $s \geq 0$.*

Proof. By Equation (3),

$$\Phi_L(s) = (1+s)^{-1/2} g(\tau), \quad \tau := \frac{a^2 s}{A} \in [0, s], \quad g(\tau) := \frac{1}{2} \sqrt{1+\tau} (1 + e^{-\tau}).$$

Now $g(0) = 1$, and $(\log g)'$ has the sign of $e^\tau - 1 - 2\tau$, which is first negative and then positive. Hence $g \leq \max\{1, g(s)\}$ on $[0, s]$, proving the claim. $\square$

Proposition 1 (Finite certificate). *For $\mu_0 = 0.45$, we have $K = 4$. At $s = s_\times$, every configuration with $2 \leq |L| \leq 4$ satisfies*

$$\Phi_L(s_\times) \leq \text{env}(s_\times).$$

Proof. In the coordinates $x_j = w_j^2$, the feasible region is

$$\mathcal{X} := \left\{ \mathbf{x} : \mu_0^2 \leq x_j \leq 1, \sum_j x_j \leq 1 \right\},$$

where $\rho^2 = 1 - \sum_j x_j \geq 0$, and Equation (3) is smooth. Every partial derivative satisfies

$$\left| \frac{\partial \Phi_L}{\partial x_j} \right| \leq \Lambda := \frac{\sqrt{s_\times/\pi}}{2\mu_0} + \frac{s_\times}{2}.$$

Indeed, $\partial_{x_j} = (2w_j)^{-1} \partial_{w_j}$ on the cosine factor, and

$$\left| \frac{\partial}{\partial w_j} \cos^2 \left(\frac{w_j \xi}{2} \right) \right| = \left| \frac{\xi}{2} \sin(w_j \xi) \right| \leq \frac{|\xi|}{2}.$$

Since $|w_j| \geq \mu_0$ and $\mathbb{E}|\xi| = 2\sqrt{s_\times/\pi}$, this contributes the first term in Λ . Differentiating $e^{-\rho^2 \xi^2 / 4}$ with respect to x_j contributes at most $\xi^2/4$. Its expectation is $s_\times/2$ because $\mathbb{E}[\xi^2] = 2s_\times$.

The computation also uses a second-order bound. Put $c_j(x_j) := \cos^2(\sqrt{x_j}\xi/2)$ and $M_r := \mathbb{E}[|\xi|^r]$ for $\xi \sim N(0, 2s_\times)$. On $\mathcal{X}$,

$$|c'_j| \leq \frac{|\xi|}{4\mu_0}, \quad |c''_j| \leq \frac{\xi^2}{8\mu_0^2} + \frac{|\xi|}{8\mu_0^3}.$$

The first and second derivatives of the Gaussian factor contribute respectively $\xi^2/4$ and $\xi^4/16$. Applying the product rule and taking expectations therefore gives

$$\begin{aligned} \left| \frac{\partial^2 \Phi_L}{\partial x_j^2} \right| &\leq \frac{M_4}{16} + \frac{M_3}{8\mu_0} + \frac{M_2}{8\mu_0^2} + \frac{M_1}{8\mu_0^3}, \\ \left| \frac{\partial^2 \Phi_L}{\partial x_j \partial x_k} \right| &\leq \frac{M_4}{16} + \frac{M_3}{8\mu_0} + \frac{M_2}{16\mu_0^2} \quad (j \neq k). \end{aligned}$$

Here

$$M_1 = 2\sqrt{\frac{s_\times}{\pi}}, \quad M_2 = 2s_\times, \quad M_3 = 2\sqrt{\frac{2}{\pi}}(2s_\times)^{3/2}, \quad M_4 = 12s_\times^2.$$

For $d = |L|$, let H_d be the diagonal bound plus $(d-1)$ times the off-diagonal bound. On a box centred at a feasible $\mathbf{c}$, let r_j bound the displacement of coordinate j . The first-order and Taylor bounds are

$$\Phi_L(\mathbf{x}) \leq \Phi_L(\mathbf{c}) + \Lambda \sum_j r_j$$

and

$$\Phi_L(\mathbf{x}) \leq \Phi_L(\mathbf{c}) + \sum_j \left| \frac{\partial \Phi_L}{\partial x_j}(\mathbf{c}) \right| r_j + \frac{H_d}{2} \left(\sum_j r_j \right)^2.$$

The accompanying notebook performs a branch-and-bound search over $\mathcal{X}$. It discards infeasible boxes, evaluates Equation (3) and its gradient at a feasible centre, takes the smaller of the two displayed upper bounds, and bisects the longest edge until every box lies below $\text{env}(s_\times)$. For $d = 2, 3, 4$, it processes respectively 645, 9,241, and 33,687 boxes. The largest accepted upper bound is approximately 0.549600938, more than $1.3 \cdot 10^{-6}$ below $\text{env}(s_\times) \approx 0.549602277$. $\square$

Proposition 2 (Uniform transform bound).

$$\sup_{\|\mathbf{w}\|=1} \mathcal{L}_{\mathbf{w}}(s_\times) \leq 0.5508.$$

Proof. Apply Lemma 16 with $\mu_0 = 0.45$ and $s = s_\times$. The bound $\Phi_L(s_\times) \leq \text{env}(s_\times)$ follows from Equation (3) when $L = \emptyset$, from Lemma 17 when $|L| = 1$, and from Proposition 1 when $2 \leq |L| \leq 4$. Moreover,

$$\Pr_{\xi \sim N(0, 2s_\times)}[|\xi| > \pi/0.45] < 2^{-9.74}.$$

Therefore

$$\mathcal{L}_{\mathbf{w}}(s_\times) < \text{env}(s_\times) + 2^{-9.74} \leq 0.5508. \quad \square$$

Proof of the lower-tail bound. Taking $s = s_\times$ in Equation (2) and applying Proposition 2 gives

$$\Pr[\|\mathbf{J}\mathbf{w}\|^2 \leq \alpha] \leq e^{\alpha s_\times} (0.5508)^{2\lambda} \leq 2^{-(\lambda+1)}$$

at the value $\alpha = \alpha(\lambda)$ in Table 1, namely

$$\alpha(\lambda) = -\frac{(\lambda+1) + 2\lambda \log_2 0.5508}{s_\times \log_2 e}.$$

$\square$

4.4 Reduction Modulo q

Reduction modulo q can make a large integral projection appear short through wrap-around. We follow the blueprint of [BS23, Appendix A] and instantiate its three regimes with the real and single-row bounds proved above.

For each of the five security levels in Table 1, let $T_\lambda, D_\lambda, r_\lambda$ be as in Table 2, and define

$$A_\lambda := r_\lambda D_\lambda, \quad C_\lambda := \frac{D_\lambda}{\sqrt{1 - r_\lambda^{-2}}}, \quad \delta_\lambda := \frac{0.52\sqrt{2}}{\sqrt{r_\lambda^2 - 1}}, \quad \tau_\lambda := \kappa_{\text{BD}} \operatorname{erfc}(D_\lambda/2).$$

Table 2: Auxiliary constants for the modular bound.

λ	T_λ	D_λ	r_λ
64	16	7.34	5.37
96	23	8.83	5.99
128	30	10.08	6.53
192	44	12.20	7.44
256	58	13.98	8.22

Writing $\ell_\lambda := \sqrt{\alpha(\lambda)}$, the values in Tables 1 and 2 satisfy²

$$2^{-2\lambda} \sum_{i=0}^{T_\lambda-1} \binom{2\lambda}{i} \leq 2^{-\lambda},$$

$$b(\lambda) > \max \left\{ \frac{\ell_\lambda}{1 - \gamma_1(\lambda)/D_\lambda}, \frac{2A_\lambda \ell_\lambda}{\sqrt{T_\lambda}}, \frac{2C_\lambda \ell_\lambda}{\sqrt{\pi} (2^{-1/2} - 2\delta_\lambda - \tau_\lambda)} \right\}. \quad (4)$$

Proof of the modular bound in Theorem 2. Fix λ , abbreviate $T = T_\lambda$, $D = D_\lambda$, $A = A_\lambda$, $C = C_\lambda$, $\delta = \delta_\lambda$, $\tau = \tau_\lambda$, $\ell = \ell_\lambda$, and $b = b(\lambda)$, and set $t := \ell\theta$.

Case 1: $\theta \leq \|\mathbf{w}\| < q/D$. If $\|\mathbf{J}\mathbf{w} \bmod q\| \leq t$, then either

$$\|\mathbf{J}\mathbf{w}\| \leq t \leq \ell\|\mathbf{w}\|,$$

or some row $\mathbf{u}_j$ satisfies

$$|\langle \mathbf{w}, \mathbf{u}_j \rangle| \geq q - t \geq q \left(1 - \frac{\ell}{b} \right) > \frac{\gamma_1 q}{D} > \gamma_1 \|\mathbf{w}\|.$$

The lower-tail bound and a union bound over the single-row estimate in Theorem 2 give

$$2^{-(\lambda+1)} + (2\lambda) \frac{2^{-(\lambda+1)}}{2\lambda} = 2^{-\lambda}.$$

Case 2: $\|\mathbf{w}\|_\infty \geq q/A$. Fix h with $|w_h| \geq q/A$. For each row, condition on all entries except $u_{j,h}$. Then

$$\Pr \left[|\langle \mathbf{w}, \mathbf{u}_j \rangle \bmod q| \geq \frac{|w_h|}{2} \geq \frac{q}{2A} \right] \geq \frac{1}{2}.$$

Indeed, if the choice $u_{j,h} = 0$ has balanced absolute value less than $|w_h|/2$, both choices $u_{j,h} = \pm 1$ have balanced absolute value at least $|w_h|/2$, since $|w_h| \leq q/2$ rules out a second

²The final cell of the notebook re-checks these inequalities at 200 bits of precision.

wrap-around. Otherwise, the choice $u_{j,h} = 0$ suffices. Thus the number of rows satisfying the display stochastically dominates $\text{Bin}(2\lambda, 1/2)$. By Equation (4),

$$t \leq \frac{\ell q}{b} < \frac{\sqrt{T}q}{2A}.$$

Hence $\|\mathbf{J}\mathbf{w} \bmod q\| \leq t$ implies that fewer than T rows satisfy the display, and its probability is at most

$$2^{-2\lambda} \sum_{i=0}^{T-1} \binom{2\lambda}{i} \leq 2^{-\lambda}.$$

Case 3: $\|\mathbf{w}\| \geq q/D$ and $\|\mathbf{w}\|_\infty < q/A$. There is a vector $\mathbf{v}$ with disjoint-support decomposition $\mathbf{w} = \mathbf{v} + (\mathbf{w} - \mathbf{v})$ such that every $v_i \in \{0, w_i\}$ and

$$\frac{q}{C} < \|\mathbf{v}\| < \frac{q}{D}.$$

To obtain $\mathbf{v}$, remove coordinates from $\mathbf{w}$ until its norm first falls below q/D . The last removal changes its squared norm by less than q^2/A^2 , and hence

$$\|\mathbf{v}\|^2 > \frac{q^2}{D^2} - \frac{q^2}{A^2} = \frac{q^2}{C^2}.$$

For a row $\mathbf{u}$, set $X := \langle \mathbf{v}, \mathbf{u} \rangle$ and $Z := \langle \mathbf{w} - \mathbf{v}, \mathbf{u} \rangle$. These variables are independent. The Berry–Esseen estimate used in [BS23, Appendix A] gives, for $Y \sim N(0, \|\mathbf{v}\|^2/2)$,

$$\sup_x |\Pr[X \leq x] - \Pr[Y \leq x]| \leq 0.52\sqrt{2} \frac{\|\mathbf{v}\|_\infty}{\|\mathbf{v}\|} < \frac{0.52\sqrt{2}}{\sqrt{r_\lambda^2 - 1}} = \delta.$$

Fix $Z \bmod q = a$. Within $[-q/2, q/2]$, the set

$$G_a := \{x \in [-q/2, q/2] : |x + a \bmod q| \leq t\}$$

has measure at most $2t$ and is either an interval or the complement of an interval. Since the density of Y is at most $1/(\sqrt{\pi}\|\mathbf{v}\|)$,

$$\Pr[X \in G_a] \leq \Pr[Y \in G_a] + 2\delta \leq \frac{2t}{\sqrt{\pi}\|\mathbf{v}\|} + 2\delta < \frac{2C\ell}{b\sqrt{\pi}} + 2\delta.$$

Furthermore, Lemma 12 and $\|\mathbf{v}\| < q/D$ give

$$\Pr[|X| > q/2] < \kappa_{\text{BD}} \text{erfc}(D/2) = \tau.$$

Consequently, for every row and every value of $Z \bmod q$,

$$\Pr[|X + Z \bmod q| \leq t] < \frac{2C\ell}{b\sqrt{\pi}} + 2\delta + \tau < 2^{-1/2},$$

where the last inequality is Equation (4). If $\|\mathbf{J}\mathbf{w} \bmod q\| \leq t$, this event holds in all 2λ independent rows. Its probability is therefore less than $(2^{-1/2})^{2\lambda} = 2^{-\lambda}$. $\square$

5 Sampling Small-Norm Challenges

Short challenges with invertible pairwise differences are crucial building blocks for a large family of folding schemes [KLOT25, BC25b, BC25a, GLLO26, NS26, Osa26, BNP26] and argument systems [BLS19, BLNS20, AL21, CLM23, LNP22, BS23, AFLN24, NS24, LSS24, KLNO25, KLOT25, Che25, BBL⁺26b, KLN⁺26, BBL⁺26a].

To see why the choice of challenges matters, consider one folding step with challenge c :

$$\mathbf{w}_{\text{fold}} = \mathbf{w}_{\text{acc}} + c\mathbf{w}, \quad \|\mathbf{w}_{\text{fold}}\| \leq \|\mathbf{w}_{\text{acc}}\| + \|c\|_{\text{op}}\|\mathbf{w}\|.$$

The multiplication operator norm of c therefore governs witness growth in the folding direction. Extraction additionally requires distinct challenges to have invertible differences. A useful challenge set must be large, have invertible pairwise differences, and contain elements with small multiplication operator norm.

The naive construction bounds every power-basis coefficient of c by B . The short-invertible-elements theorem of [LS18] makes the resulting set a strong sampling set for suitable conductors and moduli. For a power-of-two conductor, the direct expansion estimate is only $\|c\|_{\text{op}} \leq \varphi B$. Restricting the challenge to exactly s non-zero coefficients improves this estimate to sB , while retaining $(2B)^s \binom{\varphi}{s}$ possible challenges.

We further reduce the expansion factor by rejection sampling. The sampler draws uniformly from the bounded fixed-weight set and rejects any challenge whose multiplication operator norm exceeds a threshold γ . If the acceptance probability is p , the expected number of trials is p^{-1} and the accepted set loses exactly $-\log_2 p$ bits of cardinality. Since rejection only removes elements, it preserves pairwise-difference invertibility. The concrete measurements below show how the operator-norm threshold decreases as more sampling time is allowed.

The exact invertibility criterion imposes restrictive conditions on the conductor and modulus. Efficient implementations instead favour moduli for which the cyclotomic polynomial factors into low-degree polynomials and admits an incomplete NTT. We therefore also formulate a heuristic approximate-strong-sampling guarantee for such almost-splitting cyclotomics. If every slot has degree e and behaves approximately uniformly, a union bound over the φ/e slots suggests

$$\epsilon_C \approx \frac{\varphi}{eq^e}.$$

Cyclo [GLLO26] uses a similar slotwise heuristic. We support the estimate with pointwise measurements for selected quadratic-slot moduli.

The techniques of this section are not new. They were already sketched in [BS23]. Our contribution is a more thorough treatment of the concrete tradeoffs, a wider range of parametrisations, and a new heuristic for almost-splitting cyclotomic moduli.

The sampler results are independent of the folding scheme and apply to other lattice protocols with short ring challenges. Section 5.1 gives the exact fixed-weight construction, Section 5.2 gives the rejection sampler and its concrete tradeoffs, and Section 5.3 gives the heuristic almost-splitting instantiation.

5.1 Exact Bounded Fixed-Weight Sampler

The invertibility threshold in Theorem 3 is governed by the distortion of the power basis of $\mathbb{Q}(\zeta_u)$ under its canonical embedding, measured by the constant $s_1(u)$ of [LS18]. The embeddings σ and coeff of Definition 4 are defined for the conductor $\mathfrak{f}$ and the powerful basis, so we define s_1 directly for an arbitrary conductor.

Definition 29 (Power-Basis Embedding Norm). For a conductor u , let $(\sigma_{u,j})_{j \in [\varphi(u)]}$ be the embeddings of $\mathbb{Q}(\zeta_u)$ into $\mathbb{C}$. Define

$$s_1(u) := \|\mathbf{V}_u\|_{\text{op}}, \quad \mathbf{V}_u := (\sigma_{u,j}(\zeta_u^i))_{j,i \in [\varphi(u)]} \in \mathbb{C}^{\varphi(u) \times \varphi(u)}.$$

Theorem 3 ([LS18, Theorem 1.1]). For distinct primes $(p_i)_{i \in [t]}$, write

$$\mathfrak{f} = \prod_{i \in [t]} p_i^{e_i}, \quad z = \prod_{i \in [t]} p_i^{g_i}, \quad 1 \leq g_i \leq e_i.$$

Let q be a prime satisfying

$$q \equiv 1 \pmod{z}, \quad \text{ord}_{\mathfrak{f}}(q) = \mathfrak{f}/z.$$

Then there exist distinct $r_j \in \mathbb{F}_q^\times$ such that

$$\Phi_{\mathfrak{f}}(x) = \prod_{j \in [\varphi(z)]} (x^{\mathfrak{f}/z} - r_j) \pmod{q},$$

where every $x^{\mathfrak{f}/z} - r_j$ is irreducible in $\mathbb{F}_q[x]$. Moreover, every $y = \sum_{i \in [\varphi]} y_i \zeta_{\mathfrak{f}}^i \in \mathcal{R}$ satisfying

$$0 < \max_{i \in [\varphi]} |y_i| < \frac{1}{s_1(z)} q^{1/\varphi(z)}$$

has a unit image in $\mathcal{R}_q$.

Definition 30 (Bounded Fixed-Weight Challenges). For $1 \leq s \leq \varphi$ and $B \in \mathbb{N}$, define

$$\mathcal{C}_{\mathfrak{f},s,B}^{\text{fw}} := \left\{ \sum_{i \in [\varphi]} c_i \zeta_{\mathfrak{f}}^i : c_i \in [-B, B] \cap \mathbb{Z}, |\{i \in [\varphi] : c_i \neq 0\}| = s \right\}.$$

The case $B = 1$ gives fixed-weight ternary challenges.

Lemma 18 (Exact bounded fixed-weight sampler). For distinct primes $(p_i)_{i \in [t]}$, write

$$\mathfrak{f} = \prod_{i \in [t]} p_i^{e_i}, \quad z = \prod_{i \in [t]} p_i^{g_i}, \quad 1 \leq g_i \leq e_i.$$

Let q be an odd prime satisfying

$$q \equiv 1 \pmod{z}, \quad \text{ord}_{\mathfrak{f}}(q) = \mathfrak{f}/z.$$

For $1 \leq s \leq \varphi$ and $B \in \mathbb{N}$, if

$$2B < \frac{1}{s_1(z)} q^{1/\varphi(z)},$$

then $\mathcal{C}_{\mathfrak{f},s,B}^{\text{fw}}$ is a strong sampling set modulo q . Its cardinality and expansion factor satisfy

$$|\mathcal{C}_{\mathfrak{f},s,B}^{\text{fw}}| = (2B)^s \binom{\varphi}{s}, \quad \Gamma_{\mathcal{C}} \leq sB \max_{i \in [\varphi]} \|\zeta_{\mathfrak{f}}^i\|_{\text{op}}.$$

Proof. Let $c = \sum_i c_i \zeta_{\mathfrak{f}}^i$ and $c' = \sum_i c'_i \zeta_{\mathfrak{f}}^i$ be distinct elements of $\mathcal{C}_{\mathfrak{f},s,B}^{\text{fw}}$. The cardinality follows by choosing the support and then one of $2B$ non-zero values at each supported coordinate. For $y := c - c'$, the hypothesis gives

$$0 < \max_{i \in [\varphi]} |c_i - c'_i| \leq 2B < \frac{1}{s_1(z)} q^{1/\varphi(z)}.$$

Thus, Theorem 3 shows that $c - c'$ is invertible in $\mathcal{R}_q$, and the set is a strong sampling set.

For every $x \in \mathcal{R}$, the triangle inequality for operator norms gives

$$\|cx\| \leq \sum_{i \in [\varphi]} |c_i| \|\zeta_{\mathfrak{f}}^i\|_{\text{op}} \|x\| \leq sB \max_{i \in [\varphi]} \|\zeta_{\mathfrak{f}}^i\|_{\text{op}} \|x\|.$$

This proves the expansion bound. $\square$

Corollary 2 (Power-of-two cyclotomics). *Let $1 \leq s \leq \varphi$ and $B \in \mathbb{N}$, let $\mathfrak{f} = 2\varphi$, where $\varphi \geq k > 1$ are powers of two, and let q be a prime satisfying*

$$q = 2k + 1 \pmod{4k}.$$

If

$$2B < \frac{q^{1/k}}{\sqrt{k}},$$

then $\mathcal{C}_{2\varphi, s, B}^{\text{fw}}$ is a strong sampling set modulo q , and

$$\Gamma_{\mathcal{C}} \leq sB.$$

Proof. Apply Lemma 18 with $z = 2k$. For a power-of-two conductor, $s_1(2k) = \sqrt{k}$, and multiplication by every power of $\zeta_{2\varphi}$ is a signed permutation in the coefficient embedding. Hence $\max_i \|\zeta_{2\varphi}^i\|_{\text{op}} = 1$. $\square$

5.2 Operator-Norm Rejection Sampling

Definition 31 (Operator-Norm Rejection Sampler). For a finite non-empty set $\mathcal{S} \subset \mathcal{R}$ and $\gamma > 0$, define

$$\mathcal{S}(\gamma) := \{c \in \mathcal{S} : \|c\|_{\text{op}} \leq \gamma\}, \quad p_{\mathcal{S}}(\gamma) := \frac{|\mathcal{S}(\gamma)|}{|\mathcal{S}|}.$$

The sampler draws $c \leftarrow \mathcal{S}$ independently until $\|c\|_{\text{op}} \leq \gamma$, and then returns c . For a power-of-two conductor, Lemma 2 gives $\|c\|_{\text{op}} = \|c\|_{\sigma, \infty}$, so the test can be performed through the canonical evaluations of c .

Lemma 19 (Rejection tradeoff). *Suppose that $p_{\mathcal{S}}(\gamma) > 0$ and that $\mathcal{S}$ is a strong sampling set modulo q . Then the output of Definition 31 is uniform over $\mathcal{S}(\gamma)$, the expected number of trials is*

$$\frac{1}{p_{\mathcal{S}}(\gamma)},$$

and

$$\log_2 |\mathcal{S}(\gamma)| = \log_2 |\mathcal{S}| + \log_2 p_{\mathcal{S}}(\gamma).$$

The accepted set is a strong sampling set modulo q , and its expansion factor satisfies

$$\Gamma_{\mathcal{C}} \leq \gamma.$$

Proof. For every $c \in \mathcal{S}(\gamma)$,

$$\Pr[\text{the sampler returns } c] = \sum_{j \geq 1} (1 - p_{\mathcal{S}}(\gamma))^{j-1} \frac{1}{|\mathcal{S}|} = \frac{1}{|\mathcal{S}(\gamma)|}.$$

The number of trials is geometric with parameter $p_{\mathcal{S}}(\gamma)$. The cardinality identity follows from

$$|\mathcal{S}(\gamma)| = p_{\mathcal{S}}(\gamma) |\mathcal{S}|.$$

Every difference of two distinct accepted elements is invertible because $\mathcal{S}(\gamma) \subseteq \mathcal{S}$. The expansion bound follows directly from the acceptance condition. $\square$

We evaluate the concrete tradeoffs of rejection sampling experimentally. For each degree $\varphi \in \{64, 128, 256\}$, we choose the smallest coefficient bound B_{φ} for which all displayed challenge-collision targets are attainable. This gives

$$B_{64} = 2, \quad B_{128} = B_{256} = 1.$$

For a challenge-collision target $2^{-\lambda}$ and an expected-trial target T , define

$$s_{\varphi,\lambda,T} := \min \left\{ s \in \{1, \dots, \varphi\} : \left| \mathcal{C}_{2\varphi,s,B_\varphi}^{\text{fw}} \right| \geq T2^\lambda \right\}.$$

Thus, $s_{\varphi,\lambda,T}$ is the smallest weight whose base set can retain at least 2^λ elements at acceptance probability $1/T$. For $T = 1$, we set $\gamma = sB_\varphi$ and perform no rejection. Otherwise, we calibrate γ at the empirical $1/T$ quantile of $\|c\|_{\text{op}}$ for $c \leftarrow \mathcal{C}_{2\varphi,s,B_\varphi}^{\text{fw}}$. Each threshold uses 2^{18} calibration samples, while the reported expected trials and retained cardinality use an independent set of 2^{18} validation samples. We report $T \in \{1, 4, 16, 64\}$.³

Table 3: Empirical operator-norm rejection tradeoffs.

Challenge-collision target	Degree	s	B	Base cardinality	Exp. trials	Accepted cardinality	Threshold γ
2^{-100}	64	22	2	$\approx 2^{100.157}$	1.00	$\approx 2^{100.157}$	44.000
2^{-100}	64	23	2	$\approx 2^{103.026}$	4.00	$\approx 2^{101.025}$	13.602
2^{-100}	64	24	2	$\approx 2^{105.798}$	15.87	$\approx 2^{101.810}$	12.540
2^{-100}	64	25	2	$\approx 2^{108.477}$	64.05	$\approx 2^{102.475}$	11.940
2^{-100}	128	21	1	$\approx 2^{100.026}$	1.00	$\approx 2^{100.026}$	21.000
2^{-100}	128	22	1	$\approx 2^{103.308}$	4.00	$\approx 2^{101.307}$	9.259
2^{-100}	128	23	1	$\approx 2^{106.512}$	15.86	$\approx 2^{102.525}$	8.773
2^{-100}	128	23	1	$\approx 2^{106.512}$	62.56	$\approx 2^{100.545}$	8.357
2^{-100}	256	17	1	$\approx 2^{103.879}$	1.00	$\approx 2^{103.879}$	17.000
2^{-100}	256	17	1	$\approx 2^{103.879}$	3.98	$\approx 2^{101.885}$	8.688
2^{-100}	256	18	1	$\approx 2^{108.610}$	15.80	$\approx 2^{104.628}$	8.365
2^{-100}	256	18	1	$\approx 2^{108.610}$	65.44	$\approx 2^{102.578}$	8.007
2^{-128}	64	34	2	$\approx 2^{128.491}$	1.00	$\approx 2^{128.491}$	68.000
2^{-128}	64	35	2	$\approx 2^{130.269}$	4.00	$\approx 2^{128.268}$	16.937
2^{-128}	64	37	2	$\approx 2^{133.555}$	16.10	$\approx 2^{129.546}$	15.792
2^{-128}	64	38	2	$\approx 2^{135.061}$	62.86	$\approx 2^{129.087}$	15.027
2^{-128}	128	31	1	$\approx 2^{129.621}$	1.00	$\approx 2^{129.621}$	31.000
2^{-128}	128	32	1	$\approx 2^{132.221}$	3.98	$\approx 2^{130.229}$	11.218
2^{-128}	128	32	1	$\approx 2^{132.221}$	16.12	$\approx 2^{128.210}$	10.381
2^{-128}	128	33	1	$\approx 2^{134.762}$	63.88	$\approx 2^{128.765}$	10.030
2^{-128}	256	23	1	$\approx 2^{131.078}$	1.00	$\approx 2^{131.078}$	23.000
2^{-128}	256	23	1	$\approx 2^{131.078}$	3.99	$\approx 2^{129.082}$	10.164
2^{-128}	256	24	1	$\approx 2^{135.357}$	15.90	$\approx 2^{131.366}$	9.699
2^{-128}	256	24	1	$\approx 2^{135.357}$	63.08	$\approx 2^{129.378}$	9.281

The base cardinalities and accepted cardinalities without rejection in Table 3 are computed exactly and displayed with rounded exponents. The no-rejection thresholds are exact; the remaining entries are empirical estimates. In every row, the reported accepted cardinality exceeds the target, and the threshold γ is an upper bound on the expansion factor of the accepted set. For example, at degree 128 and target 2^{-100} , allowing about 63 trials reduces this bound from 21 to 8.357.

Remark 6 (Fixed Versus Bounded Hamming Weight). The coefficient bound B in Table 3 should not be confused with the Hamming weight s . One could replace the fixed-weight set by

$$\mathcal{C}_{f,\leq s,B}^{\text{fw}} := \bigcup_{j=1}^s \mathcal{C}_{f,j,B}^{\text{fw}}.$$

³The calibration and validation runs behind Table 3 are https://github.com/osdnk/pikku/blob/main/scripts/operator_norm_rejection.sage.

When s lies below the largest fixed-weight layer, the summands $(2B)^j \binom{\varphi}{j}$ grow rapidly with j . Consequently, the bounded-weight union is dominated by its weight- s layer and offers little additional entropy or reduction in operator norm. For example, at $(f, s, B) = (256, 23, 1)$,

$$\log_2 |\mathcal{C}_{256,23,1}^{\text{fw}}| \approx 106.512, \quad \log_2 |\mathcal{C}_{256,\leq 23,1}^{\text{fw}}| \approx 106.677.$$

Thus, allowing all smaller weights gains only 0.165 bits. A uniform element of the bounded-weight set has weight 23 with probability approximately 0.892, weight 22 with probability approximately 0.097, and weight at most 21 with probability approximately 0.011. Using the same empirical procedure with expected-trial target 64 gives $\gamma \approx 8.317$ for bounded weight and $\gamma \approx 8.357$ for fixed weight. At the latter threshold, bounded-weight sampling takes approximately 55.8 trials instead of 62.6.

Fixed weight is also easier to sample exactly. A Fisher–Yates shuffle [FY38, Dur64] selects s distinct positions uniformly, after which independent samples from $\{\pm 1, \dots, \pm B\}$ assign their non-zero coefficients. Exact sampling from the bounded-weight union must first sample j with probability proportional to $(2B)^j \binom{\varphi}{j}$ and then invoke a weight- j sampler. Sampling a full ternary vector and rejecting on its Hamming weight is much less efficient. In degree 128, the probability that a uniform ternary vector has weight at most 23 is approximately $2^{-96.2}$. Fixed weight therefore gives a simpler exact sampler with essentially the same entropy and operator-norm tradeoff in the parameter regime of interest.

5.3 Sampling Over Almost-Splitting Cyclotomics

The exact criterion in Theorem 3 restricts the conductor and modulus. These restrictions can preclude moduli for which the cyclotomic polynomial admits an efficient incomplete NTT. We therefore consider a power-of-two conductor $f = 2\varphi$ and moduli for which $x^\varphi + 1$ factors into distinct irreducible polynomials of degree e . This is the setting our implementation instantiates, so it is the only one the concrete parameters of Section 6 need. It is also the setting in which the prediction can be checked directly. The predicted point probability lies far below any measurable frequency at the implementation’s moduli, but it remains observable at small almost-splitting primes. No corresponding heuristic claim is made for $B > 1$ or non-power-of-two conductors.

Heuristic 1 (Almost-Splitting Cyclotomic Sampler). *Let $f = 2\varphi$ be a power of two, and let q be an odd prime such that $x^\varphi + 1$ is a product of φ/e distinct irreducible factors of degree e over $\mathbb{F}_q$. Let $s \in \{1, \dots, \varphi\}$ and $\gamma > 0$, and set*

$$\mathcal{C} := \mathcal{C}_{2\varphi,s,1}^{\text{fw}}(\gamma).$$

If $|\mathcal{C}| \geq q^e$, then we conjecture that $\mathcal{C}$ is an $\epsilon_{\mathcal{C}}$ -approximate strong sampling set modulo q with

$$\epsilon_{\mathcal{C}} \approx \frac{\varphi}{eq^e},$$

and expansion factor $\Gamma_{\mathcal{C}} \leq \gamma$.

The cardinality condition $|\mathcal{C}| \geq q^e$ ensures that the challenge set has enough entropy for a slot image of size q^e to be approximately uniform. It is necessary for the intended heuristic, although it does not by itself establish uniformity.

The heuristic is motivated by the NTT-coordinate analysis of [ALS20, Section 3 and Table 1]. For independently sampled uniform ternary coefficients in a fully splitting power-of-two cyclotomic ring, the bound in that table approaches q^{-1} for the maximum probability of any value of one NTT coordinate. Extending this behaviour to a degree- e slot gives a maximum point probability of approximately q^{-e} . Fix $c_0 \in \mathcal{C}$. The difference

$c - c_0$ is a non-unit precisely when c agrees with c_0 in at least one of the φ/e slots. If every slot value of $c \leftarrow \mathcal{C}$ has probability at most approximately q^{-e} , a union bound gives

$$\epsilon_{\mathcal{C}} \approx \frac{\varphi}{eq^e}.$$

The well-spreadness bounds of [BL25, Lemmas 33 and 34] give further theoretical support for this calculation in the power-of-two case. In particular, the latter lemma treats a fixed-Hamming-weight ternary distribution in which every coefficient class modulo e has a prescribed weight. The analysis in [GLLO26, Appendix B] applies the same slotwise argument to approximate strong sampling with independently sampled ternary coefficients. These results do not directly cover our distribution, which fixes only the global Hamming weight and then conditions on the operator-norm bound. The first condition introduces dependencies between coefficient classes, while the second may alter point probabilities. Proving that both conditions preserve the q^{-e} per-slot scale remains an open problem.

Table 4 reports pointwise measurements for selected almost-splitting primes and weights. In these experiments, every slot has degree $e = 2$. We first sample 2^{24} challenges and build the histogram of every quadratic NTT slot. For an anchor c_0 , let $\widehat{s}(c_0)$ be the sum, over all slots, of the observed histogram mass at the slot value of c_0 . We sample 2^{24} anchors, keep the 16 anchors with largest $\widehat{s}(c_0)$, and for each of them directly estimate $\Pr_c[c \neq c_0 \wedge c - c_0 \notin \mathcal{R}_q^\times]$ on a fresh batch of 2^{24} accepted challenges. The table reports the largest direct estimate among these anchors.⁴

Table 4: Observed pointwise non-unit frequencies for selected quadratic-slot moduli.

q	s	$ \mathcal{C}_{256,s,1}^{\text{fw}} $	γ	Expected trials	$ \mathcal{C}_{256,s,1}^{\text{fw}}(\gamma) $	$\varphi/(eq^e)$	Observed $\epsilon_{\mathcal{C}}$
127	3	$\approx 2^{21.381}$	2.9	5.04	$\approx 2^{19.047}$	$\approx 2^{-7.977}$	$\approx 2^{-7.959}$
383	4	$\approx 2^{27.347}$	3.4	9.86	$\approx 2^{24.045}$	$\approx 2^{-11.162}$	$\approx 2^{-10.928}$
1151	4	$\approx 2^{27.347}$	3.7	2.92	$\approx 2^{25.802}$	$\approx 2^{-14.337}$	$\approx 2^{-14.089}$
1279	4	$\approx 2^{27.347}$	3.7	2.92	$\approx 2^{25.802}$	$\approx 2^{-14.642}$	$\approx 2^{-14.463}$
1663	4	$\approx 2^{27.347}$	3.8	1.71	$\approx 2^{26.572}$	$\approx 2^{-15.399}$	$\approx 2^{-15.286}$
2687	5	$\approx 2^{32.979}$	4.0	8.35	$\approx 2^{29.918}$	$\approx 2^{-16.784}$	$\approx 2^{-16.524}$
5119	5	$\approx 2^{32.979}$	4.0	8.35	$\approx 2^{29.918}$	$\approx 2^{-18.643}$	$\approx 2^{-18.385}$
6143	5	$\approx 2^{32.979}$	4.1	5.61	$\approx 2^{30.491}$	$\approx 2^{-19.169}$	$\approx 2^{-18.956}$

6 Parameters and Concrete Performance

We now evaluate the concrete performance of **PikkuFold**. We estimate the communication per fold in Section 6.1, report implementation runtimes in Section 6.2, and compare our scheme with other lattice-based folding schemes in Section 6.3.

6.1 Estimated Communication

Table 5 estimates the communication per fold with $k = 2$ across three cyclotomic degrees. Each witness has $M = \varphi m$ coordinates over $\mathbb{Z}_q$, sampled uniformly from $\{-2^{10}, \dots, 2^{10}\}$. We use security level $\lambda = 128^5$ and failure target $\kappa = 2^{-100}$. The estimates use the prime $q = 2^{64} - 59$, the 64-trial sampler parameters of Table 3, compact centred encodings for the short projection image, and commitment ranks accounting for 32 sequential folding

⁴The sampling experiment is <https://github.com/osdnk/pikku/tree/main/approximate-strong-sampling>, and the remaining columns of Table 4 are computed by https://github.com/osdnk/pikku/blob/main/scripts/almost_splitting_table.sage.

⁵We estimate the security level using the Lattice Estimator [APS15, Commit 53da598].

rounds. We use average-case bounds to choose the witness and honest-accumulator caps. For extraction, we apply Remark 3.

The k fresh commitments are reported separately from the in-protocol communication, and the accumulator commitment is not retransmitted.⁶

Table 5: Estimated communication in KB for one fold with $k = 2$ and $M = \varphi m$ coordinates over $\mathbb{Z}_q$. “Comm.” counts the k fresh commitments, and “Protocol” counts the in-protocol communication.

Degree φ	$M = 2^{25}$		$M = 2^{27}$		$M = 2^{29}$	
	Comm.	Protocol	Comm.	Protocol	Comm.	Protocol
$\varphi = 64$	18.41	4.80	18.41	4.92	19.38	5.04
$\varphi = 128$	17.44	7.16	19.38	7.28	19.38	7.40
$\varphi = 256$	19.38	11.97	19.38	12.09	19.38	12.21

6.2 Implementation

We measure one fold at the parameters of the preceding sections: $\mathcal{R}$ of conductor 256 with $\varphi = 128$, the almost-splitting prime $q = 2^{50} - 2687$ with $e = a = 2$, $k = 2$ fresh inputs and one accumulator, witness coefficients sampled uniformly from $\{-2^{10}, \dots, 2^{10}\}$, fold challenges from $\mathcal{C}_{256,23,1}^{\text{fw}}(8.357)$ sampled under Heuristic 1, which raises the knowledge error to $\kappa \approx 2^{-90}$, three projection layers of 256 rows each with $\mu = 2$ trace-batching points, and a commitment key of rank n chosen for 32 sequential folds. The two coarse layer widths split the shrinkage from km ring elements to 256 as evenly as possible.

Our implementation⁷ extends the codebase of [KLN⁺26] and extensively uses AVX-512 vectorisation. RoKoko [KLN⁺26] fixes the arithmetic modulus at 50 bits and our parameter selection reflects this constraint. The protocol is non-interactive via Fiat–Shamir. All parties run single-threaded on one core of an Intel i7-11850H. Table 6 reports the runtimes of one fold. Commitment time is reported per fresh input and excludes expansion of the commitment key from a public seed. The prover runs the witness rounds of the batched ring sumcheck over $\mathbb{F}_{q^2}$. The projections take about half of the prover time, and the sumcheck about a quarter at every size. Verifier time is dominated by the terminal evaluation of the layer descriptions and the expansion of the projection matrices from the transcript, both of which scan the S_{JL} ternary entries.

The communication row of Table 6 counts every folding message, with each coefficient encoded using its centred representative. It excludes the fresh input commitments, which occupy a further 18 to 20 KB and dominate the total. Each round polynomial is encoded by two $\mathbb{F}_{q^2}$ coefficients, with the third determined by the sumcheck consistency equation. The terminal message consists of the $k + 1$ evaluations t'_i together with the μ batched projections and $\mathbf{v}_{\text{tr}}$, whose entries are short and cost about 30 bits each.

⁶The estimates come from <https://github.com/osdnk/pikku/blob/main/estimates.ipynb>.

⁷Available at <https://github.com/osdnk/pikku/tree/main/pikku-fold-schemelet>.

Table 6: Runtimes and communication of one fold. The lower block reports SALSAA [KLOT25] on the same machine at matched input sizes. Its largest instance exhausted the machine’s memory.

Input dimensions (k, m, φ)	$(2, 2^{18}, 128)$	$(2, 2^{20}, 128)$	$(2, 2^{22}, 128)$
Commitment rank n	12	12	13
Sumcheck rounds ℓ	40	43	46
Commitment, per input	191 ms	803 ms	2280 ms
Prover	455 ms	1699 ms	6539 ms
Verifier	3.87 ms	5.39 ms	9.80 ms
Communication	5.62 KB	5.72 KB	5.83 KB
SALSAA [KLOT25], matched input sizes			
Commitment, per input	413 ms	1806 ms	—
Prover	7662 ms	31668 ms	—
Verifier	0.42 ms	0.43 ms	—
Communication	62.25 KB	66.06 KB	—

6.3 Comparison with Other Lattice-Based Folding Schemes

For the comparison with similarly sized SALSAA [KLOT25] instances in Table 6, we ran its implementation on the same benchmark machine and matched the input sizes measured in $\mathbb{Z}_q$ -elements.

Table 7 compares the communication of one folding step across lattice-based folding schemes. Each figure counts every message the prover sends during the fold, including any commitments to decomposed or otherwise transformed witnesses sent along the way, and excludes the incoming instances themselves. The values for LatticeFold+ [BC25b], the scheme of [Osa26], ProtogaLattice [BNP26] and Cyclo [GLLO26] are the estimates reported in those works, and the SALSAA [KLOT25] value is measured by its implementation. The ProtogaLattice figure sums its folding and witness-bootstrapping messages, which together form one step, and excludes the range proofs its relation additionally requires. LatticeFold [BC25a] and Neo [NS26] report no concrete communication, so we assemble it from the prover messages of their protocols at the parameters proposed there. The figures are indicative rather than directly comparable: the moduli range from 50 to 128 bits, the witness sizes, folding arities and soundness targets differ, and the folded relations range from full constraint systems [BC25a, NS26] to committed evaluation claims.

The folding messages of `PikkuFold` occupy 5.7 KB (for the parameters of Section 6.2) against 31.8 KB for Cyclo, the smallest figure among the prior schemes. The difference lies mostly in the in-protocol commitments. Every prior scheme in Table 7 commits to decomposed or transformed witnesses during the fold, and Cyclo’s extension commitment alone accounts for about 10 KB of its total. In contrast, `PikkuFold` sends no commitment beyond those of the fresh inputs. The advantage is also not an artefact of the smaller modulus, since the 64-bit modulus estimates in Table 5 give folding messages of 7.3 KB for the same ring degree and approximately 5 KB at ring degree 64.

Table 7: Communication of one folding step, excluding the incoming instances' commitments.

	Fresh inputs	$\log_2 q$	Coordinates per input	Communication
LatticeFold [BC25a]	1	64	2^{32}	≈ 250 KB
LatticeFold+ [BC25b]	1	128	2^{27}	$\lesssim 100$ KB
Neo [NS26]	1	64	2^{26}	≈ 140 KB
Scheme of [Osa26]	1	64	2^{27}	≈ 110 KB
ProtogaLattice [BNP26]	1	64	2^{31}	≈ 83 KB
SALSAA [KLOT25]	4	50	2^{26}	62.25 KB
Cyclo [GLLO26]	1	50	2^{27}	31.8 KB
PikkuFold	2	50	2^{27}	5.72 KB

7 Extensions

PikkuFold operates on the principal relation Ξ^{lin} and increases the accumulator norm at every step. This section explains two extensions. First, an occasional exact norm check and decomposition reset the norm budget, so the number of folds is no longer limited by additive norm growth. Second, the AIR reduction of [KLOT25] produces precisely the polynomial-evaluation claims accepted by Π^{fold} (cf. Figure 2).

7.1 Periodic Norm Reset

The relation Ξ^{lin} is a one-column principal linear relation of the form used in [KLNO24, KLNO25, KLOT25]. Indeed, its two algebraic claims can be written as the single linear system

$$\begin{pmatrix} \mathbf{F} \\ \widetilde{\mathbf{eq}}(\mathbf{s})^\top \end{pmatrix} \mathbf{w} = \begin{pmatrix} \mathbf{y} \\ t \end{pmatrix} \pmod{q}.$$

The exact norm-check and decomposition reductions of [KLNO24, KLOT25] therefore apply to Ξ^{lin} .

Suppose that the accumulator immediately after a checkpoint has norm bound β_{reset} . After T invocations that each fold k fresh witnesses of norm at most β_{in} , Theorem 1 gives

$$\beta_T = \beta_{\text{reset}} + Tk\Gamma_C\beta_{\text{in}}.$$

This is only the correctness-side budget. For extraction, let β'_{reset} be the bound restored by the exact norm check at the end of the interval. In the present one-column relation, $\beta'_{\text{reset}} = \beta_{\text{cap}}$. Let β'_{cap} be the maximum extraction bound allowed between checkpoints. Recalling Theorem 1, set

$$\beta'_{\text{in}} := \sqrt{k}\beta_{\text{in}} \prod_{j \in [d]} \frac{u_j}{\ell_j}.$$

Unwinding t ordinary folds from the next exact norm check gives

$$\beta'_t = \beta'_{\text{reset}} + tk\Gamma_C\beta'_{\text{in}}.$$

Consequently, both budgets are respected when

$$T \leq \min \left\{ \left\lfloor \frac{\beta_{\text{cap}} - \beta_{\text{reset}}}{k\Gamma_C\beta_{\text{in}}} \right\rfloor, \left\lfloor \frac{\beta'_{\text{cap}} - \beta'_{\text{reset}}}{k\Gamma_C\beta'_{\text{in}}} \right\rfloor \right\}. \quad (5)$$

Let β^* be the largest norm at which SIS for $\mathbf{F}$ is assumed hard. For folding arity a , define

$$\beta_{\text{sis}}^{(a)}(B) := 6\Gamma_C B.$$

The extraction cap must then satisfy

$$\beta_{\text{sis}}^{(k)}(\beta'_{\text{cap}}) < \beta^*.$$

At a checkpoint, the exact norm check of [KLOT25] enforces $\|\mathbf{w}\| \leq \beta_{\text{cap}}$ and replaces the existing evaluation claim by one at a new verifier-chosen point $\mathbf{r}_{\text{check}}$, while preserving $\mathbf{F}$, $\mathbf{y}$, and $\mathbf{w}$. Write $t_{\text{check}} := \text{MLE}[\mathbf{w}](\mathbf{r}_{\text{check}}) \bmod q$.

The prover then decomposes $\mathbf{w}$ in a centred base b , using the decomposition reduction of [KLNO24]. For L digits, write

$$\mathbf{w} = \sum_{j \in [L]} b^j \mathbf{w}_j, \quad \|\mathbf{w}_j\| \leq \beta_{\text{dig}}.$$

The output commitment and evaluation claims decompose with the same coefficients. Defining

$$\mathbf{y}_j := \mathbf{F} \mathbf{w}_j \bmod q, \quad t_j := \text{MLE}[\mathbf{w}_j](\mathbf{r}_{\text{check}}) \bmod q,$$

gives the public recombination checks

$$\mathbf{y} = \sum_{j \in [L]} b^j \mathbf{y}_j \pmod{q}, \quad t_{\text{check}} = \sum_{j \in [L]} b^j t_j \pmod{q}.$$

Thus, every digit is a fresh instance $((\mathbf{F}, \mathbf{y}_j, \mathbf{r}_{\text{check}}, t_j), \mathbf{w}_j) \in \Xi^{\text{lin}}$ under the original commitment key and a common evaluation point.

Let $K \geq L$ be an arity supported by Theorem 1, padding with zero witnesses as needed. With a zero accumulator, the checkpoint is

$$\Xi_{\beta_{\text{cap}}}^{\text{lin}} \xrightarrow{\Pi^{\text{norm}}} \Xi_{\beta_{\text{cap}}}^{\text{lin}} \xrightarrow{\Pi^{\text{dec}}} (\Xi_{\beta_{\text{dig}}}^{\text{lin}})^K \xrightarrow{\Pi^{\text{fold}}} \Xi_{K\Gamma_C \beta_{\text{dig}}}^{\text{lin}}.$$

Here, Π^{norm} and Π^{dec} denote the cited reductions. If $K\Gamma_C \beta_{\text{dig}} < \beta_{\text{cap}}$ and $\beta_{\text{sis}}^{(K)}(\beta'_{\text{cap}}) < \beta^*$, the checkpoint can be repeated indefinitely. Its relatively high cost is amortised over the intervening ordinary folds of **PikkuFold**.

7.2 Folding AIR Relations

Another apparent limitation of **PikkuFold** is that it only folds the principal relation Ξ^{lin} . Here, we argue that this relation is exactly what is needed to fold the witnesses of an AIR relation. The Algebraic Intermediate Representation (AIR) [BBHR18b] originated from the line of work on hash-based STARKs [BBHR18a, Sta21]. AIR is particularly useful when verifier succinctness is a priority, as the instance description is as small as the description of the transition function. In practice, this arithmetic relation is often used to encode the transition function of a RAM program. The construction of [BBHR18b] already does so, giving an AIR for the DNA profile match (DPM) program, and later STARK-friendly architectures use the same encoding, among them Cairo [GPR21] and the RISC-V zkVM of RISC Zero [BGt23].

Definition 32 (AIR Relation). Following [KLOT25], let $N, t, D \in \mathbb{N}$ with N and t powers of two. An instance consists of a transition polynomial

$$f \in \mathcal{R}_q[y_0, \dots, y_{2t-1}]^{\leq D}$$

and boundary values $C \subseteq [N] \times [t] \times \mathcal{R}_q$. A witness is a trace $\mathbf{W} \in \mathcal{R}^{N \times t}$. We define $((f, C), \mathbf{W}) \in \Xi_{N,t,D}^{\text{AIR}}$ if

$$\begin{aligned} f(\mathbf{W}_{i,:}, \mathbf{W}_{i+1,:}) &= 0 \pmod{q} && \text{for every } i \in [N-1], \\ \mathbf{W}_{i,j} &= u \pmod{q} && \text{for every } (i, j, u) \in C. \end{aligned}$$

We refer to [KLOT25] for the details of the reduction. It produces $2t$ fresh instances of Ξ^{lin} , ignoring the impact of the decomposition necessary to commit to the witnesses:

$$((\mathbf{F}, \mathbf{y}_j, \mathbf{r}, t_j), \mathbf{w}_j) \in \Xi_{\mathcal{R}, q, N, n, \beta}^{\text{lin}} \quad \text{for } j \in [2t],$$

at a common evaluation point and committed under the same commitment key. Set the folding arity to $k = 2t$. They can be folded with the current accumulator as

$$(\Xi_{\mathcal{R}, q, N, n, \beta}^{\text{lin}})^k \times \Xi_{\mathcal{R}, q, N, n, \beta_{\text{acc}}}^{\text{lin}} \longrightarrow \Xi_{\mathcal{R}, q, N, n, \beta_{\text{acc}} + k\Gamma_C \beta}^{\text{lin}},$$

yielding a folding scheme for AIR relations.

Remark 7 (Sumcheck Batching). As observed in [KLOT25, Section 8.1], the sumcheck for the AIR reduction can be batched with the sumcheck of `PikkuFold` in Figure 2.

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A Eliminating the Exponential Loss in Extraction

The SIS-break norm β_{sis} of Theorem 1 contains the factor $(2\Gamma_C)^{2k}$. This appendix removes it. The factor comes from one step of the knowledge-soundness proof, the stage *Binding two independent extractions* (the binding stage), where two independently extracted families are compared after all their challenge denominators have been multiplied out. If the

two families are instead obtained at the same folding challenge, they can be compared coordinate by coordinate. Each comparison is then a sum of three products of a challenge difference with a single witness, instead of a product over $2k$ differences, and the norm drops to $6\Gamma_C\beta'_{\text{out}}$. The price is a more careful extractor, because the common challenge is taken from a run that happened to succeed, and everything sampled afterwards is biased by that selection.

The idea and probabilistic argument come from [WF26], where they were developed for Cyclo [GLLO26].

A.1 Exponential loss in naive extraction

Recall the binding stage. A coordinate-wise forking extraction returns a baseline witness $\mathbf{u}^{(k)}$ and, for each $i \in [k]$, a fork witness $\mathbf{u}^{(i)}$ obtained by changing only the i th folding coordinate. Their differences satisfy $\mathbf{F}\mathbf{a}_i = \Delta_i \mathbf{y}_i$, so $\mathbf{w}_i^* = \Delta_i^{-1} \mathbf{a}_i$ opens $\mathbf{y}_i$ modulo q . Nothing bounds $\|\Delta_i^{-1}\|_{\text{op}}$, so $\mathbf{w}_i^*$ has no useful norm bound. The proof of Theorem 1 therefore multiplies all denominators out and works with $\mathbf{a}_{\text{acc}} = \Delta \mathbf{u}^{(k)} - \sum_i z_i^{(k)} \Delta_{-i} \mathbf{a}_i$, an integral vector that opens $\Delta \mathbf{y}_{\text{acc}}$. Its norm is at most $\beta_{\text{bind}} = (k+1)\beta'_{\text{out}}(2\Gamma_C)^k$, one factor $2\Gamma_C$ per denominator. A second family, extracted using independent coins, gives $\hat{\mathbf{a}}_{\text{acc}}$ and $\hat{\Delta}$ of the same shape. The two are compared through $\mathbf{x}_{\text{acc}} = \mathbf{a}_{\text{acc}} \hat{\Delta} - \hat{\mathbf{a}}_{\text{acc}} \Delta$, which costs each family's product of denominators once more. Hence

$$\|\mathbf{x}_{\text{acc}}\| \leq 2\beta_{\text{bind}}\rho_{\text{bind}} = 2(k+1)\beta'_{\text{out}}(2\Gamma_C)^{2k}.$$

This term dominates β_{sis} .

The exponent does not appear in the per-coordinate comparison $\mathbf{x}_i = \mathbf{a}_i \hat{\Delta}_i - \hat{\mathbf{a}}_i \Delta_i$ of the same stage. That vector has norm at most $2 \cdot (2\beta'_{\text{out}}) \cdot (2\Gamma_C) = 8\Gamma_C\beta'_{\text{out}}$, because it involves only the two differences of coordinate i . The accumulator is the exception because it has no fork of its own. Its opening is what remains of the baseline after the k fresh openings are subtracted, and when the two families sit at different folding challenges $\mathbf{z} \neq \hat{\mathbf{z}}$ their baselines open different images, so the subtraction can only be compared after the denominators are cleared. If both families are held at one challenge, the baselines open the same image, the fresh openings can be compared one at a time, and the accumulator opening is determined by the rest. Appendix A.2 does the algebra and Appendix A.3 shows how to obtain two families at one challenge.

A.2 Comparison at a common challenge

The following lemma is deterministic. It only uses the statement and the folding challenge, which both families share. Every other verifier coin, every sumcheck message, and every terminal claim may differ between the two.

Lemma 20 (Comparison at a Common Challenge, Adapted from [WF26, Lemmas 1 and 2]). *Fix a statement with commitment key $\mathbf{F}$ and images $\mathbf{y}_{\text{acc}}, \mathbf{y}_0, \dots, \mathbf{y}_{k-1}$, and a folding challenge $\mathbf{z} \in \mathcal{C}^k$. Let family 0 supply integral vectors $\mathbf{u}, \mathbf{v}_0, \dots, \mathbf{v}_{k-1} \in \mathcal{R}^m$ and fork values $\zeta_0, \dots, \zeta_{k-1} \in \mathcal{C}$, and let family 1 supply $\hat{\mathbf{u}}, (\hat{\mathbf{v}}_i)_{i \in [k]}$ and $(\hat{\zeta}_i)_{i \in [k]}$ of the same type, satisfying, for every $i \in [k]$,*

$$\mathbf{F}\mathbf{u} = \mathbf{F}\hat{\mathbf{u}} = \mathbf{y}_{\text{acc}} + \sum_{j \in [k]} z_j \mathbf{y}_j \bmod q, \quad \mathbf{F}\mathbf{v}_i = \mathbf{y}_{\text{acc}} + \sum_{j \in [k] \setminus \{i\}} z_j \mathbf{y}_j + \zeta_i \mathbf{y}_i \bmod q,$$

the same equation for $\hat{\mathbf{v}}_i$ with $\hat{\zeta}_i$ in place of ζ_i , and $\|\mathbf{u}\|, \|\hat{\mathbf{u}}\|, \|\mathbf{v}_i\|, \|\hat{\mathbf{v}}_i\| \leq \beta'_{\text{out}}$. Put $\mathbf{a}_i := \mathbf{v}_i - \mathbf{u}$, $\hat{\mathbf{a}}_i := \hat{\mathbf{v}}_i - \hat{\mathbf{u}}$, $\Delta_i := \zeta_i - z_i$, and $\hat{\Delta}_i := \hat{\zeta}_i - z_i$, and assume that every Δ_i and $\hat{\Delta}_i$ is invertible modulo q . Subtracting the displayed equations gives $\mathbf{F}\mathbf{a}_i = \Delta_i \mathbf{y}_i \bmod q$ and

$\mathbf{F}\hat{\mathbf{a}}_i = \hat{\Delta}_i \mathbf{y}_i \bmod q$, and Δ_i , $\hat{\Delta}_i$ and $\hat{\Delta}_i - \Delta_i = \hat{\zeta}_i - \zeta_i$ are differences of two elements of $\mathcal{C}$, of operator norm at most $2\Gamma_{\mathcal{C}}$. Define $\mathbf{w}_i^* := \Delta_i^{-1} \mathbf{a}_i \bmod q$ and $\hat{\mathbf{w}}_i^* := \hat{\Delta}_i^{-1} \hat{\mathbf{a}}_i \bmod q$. Then exactly one of the following holds.

- (i) $\mathbf{u} \neq \hat{\mathbf{u}}$, and $\mathbf{x}_{\mathbf{u}} := \mathbf{u} - \hat{\mathbf{u}}$ satisfies $(\mathbf{F}, \mathbf{x}_{\mathbf{u}}) \in \Xi_{\mathcal{R}, q, m, n, 2\beta'_{\text{out}}}^{\text{SIS}}$.
- (ii) $\mathbf{u} = \hat{\mathbf{u}}$ and $\mathbf{w}_i^* \neq \hat{\mathbf{w}}_i^* \bmod q$ for some $i \in [k]$, and $\mathbf{x}_i := \mathbf{a}_i \hat{\Delta}_i - \hat{\mathbf{a}}_i \Delta_i$ satisfies $(\mathbf{F}, \mathbf{x}_i) \in \Xi_{\mathcal{R}, q, m, n, 6\Gamma_{\mathcal{C}}\beta'_{\text{out}}}^{\text{SIS}}$.
- (iii) The two families define the same tuple $(\mathbf{u}, (\mathbf{w}_i^*)_{i \in [k]})$, and the accumulator opening forced by linearity,

$$\mathbf{w}_{\text{acc}}^* := \mathbf{u} - \sum_{i \in [k]} z_i \mathbf{w}_i^* \bmod q,$$

satisfies $\mathbf{F}\mathbf{w}_{\text{acc}}^* = \mathbf{y}_{\text{acc}} \bmod q$ and is common to both families.

Proof. In case (i), subtracting the two baseline equations gives $\mathbf{F}\mathbf{x}_{\mathbf{u}} = 0 \bmod q$, the vector is non-zero over $\mathcal{R}$ by hypothesis, and the triangle inequality gives $\|\mathbf{x}_{\mathbf{u}}\| \leq 2\beta'_{\text{out}}$.

In case (ii), subtracting $\mathbf{F}\mathbf{a}_i \hat{\Delta}_i = \Delta_i \hat{\Delta}_i \mathbf{y}_i$ and $\mathbf{F}\hat{\mathbf{a}}_i \Delta_i = \hat{\Delta}_i \Delta_i \mathbf{y}_i$ gives $\mathbf{F}\mathbf{x}_i = 0 \bmod q$, using commutativity of $\mathcal{R}_q$. Substituting $\mathbf{a}_i = \mathbf{w}_i^* \Delta_i$ and $\hat{\mathbf{a}}_i = \hat{\mathbf{w}}_i^* \hat{\Delta}_i$ modulo q gives

$$\mathbf{x}_i = (\mathbf{w}_i^* - \hat{\mathbf{w}}_i^*) \Delta_i \hat{\Delta}_i \bmod q,$$

and multiplication by the unit $\Delta_i \hat{\Delta}_i$ is injective, so $\mathbf{x}_i \neq 0 \bmod q$ and in particular $\mathbf{x}_i \neq 0$ over $\mathcal{R}$. For the norm, the shared baseline collapses two of the four products: since $\mathbf{u} = \hat{\mathbf{u}}$,

$$\mathbf{x}_i = (\mathbf{v}_i - \mathbf{u}) \hat{\Delta}_i - (\hat{\mathbf{v}}_i - \mathbf{u}) \Delta_i = \mathbf{v}_i \hat{\Delta}_i - \hat{\mathbf{v}}_i \Delta_i - \mathbf{u} (\hat{\zeta}_i - \zeta_i),$$

three products of a challenge difference with a single witness, so $\|\mathbf{x}_i\| \leq 3 \cdot 2\Gamma_{\mathcal{C}} \cdot \beta'_{\text{out}} = 6\Gamma_{\mathcal{C}}\beta'_{\text{out}}$.

Otherwise the baselines and all normalised coordinates agree, so the reconstruction $\mathbf{w}_{\text{acc}}^*$ is the same in both families, and $\mathbf{F}\mathbf{w}_{\text{acc}}^* = \mathbf{F}\mathbf{u} - \sum_i z_i \mathbf{y}_i = \mathbf{y}_{\text{acc}} \bmod q$ by linearity. $\square$

Shortness of $\mathbf{w}_{\text{acc}}^*$ is established at the end of the extraction, once the fresh openings are known to be short, as in the additive-reconstruction stage of the proof of Theorem 1. Two properties of Π^{fold} make it easy to obtain the data of the lemma. The images $\mathbf{y}_{\text{acc}}, \mathbf{y}_0, \dots, \mathbf{y}_{k-1}$ are part of the statement, so any two accepting executions share them. The folding challenge $\mathbf{z}$ is the last verifier message, so the extractor, playing the verifier, can send a stored value of $\mathbf{z}$ to a fresh execution, and it can also re-send a stored $\mathbf{z}$ with one coordinate changed while every earlier message is kept.

A.3 Retrying at a stored challenge

The extractor must produce two successful executions at the same folding challenge. It obtains the first by running the prover once and storing the challenge. It obtains the second by re-sending the stored challenge to fresh executions until one succeeds. The stored challenge is not a uniform sample from $\mathcal{C}^k$. It was kept because the prover succeeded at it, so challenges at which the prover succeeds more often are stored more often. The coins of the second execution are selected in the same way, since the loop keeps the first success. The bounds used later in the proof of Proposition 3, on the sumcheck, the trace batching, and the projection, are stated for uniform coins. Conditioning them on success would multiply each by the inverse success probability, which is unbounded.

A second difficulty arises in the retry condition. It is tempting to let the retry loop wait for a success in which the new challenge coordinate also differs from the stored one by a unit, so that non-unit differences never occur. The bad-event bounds would then be

divided by the probability of success and a unit difference, and the ratio of that to the plain success probability is again unbounded.

Both problems have the same resolution. Every loop selects on success alone. Every bad event is charged with the probability that a fresh execution succeeds and is bad, before any selection. The number of retries is counted in expectation over the entry into the loop, not conditionally on it. With this accounting the bias cancels exactly: a challenge is stored with probability proportional to the success probability at it, and the expected number of retries at it is the inverse of the same quantity. The next two lemmas prove this for a loop that repeats all continuation coins at a fixed challenge, and for a loop that repeats one coordinate of the challenge. Both are stated in the form used by Proposition 3.

Lemma 21 (Resampling at a Stored Challenge, Adapted from [WF26, Lemmas 3 and 10]). *Let μ be a distribution on a finite set A , let ν be a distribution on a finite set B , and let $I: A \times B \rightarrow \{0, 1\}$. For $a \in A$ put $q(a) := \Pr_{b \leftarrow \nu}[I(a, b) = 1]$. Let X be a finite set of records and let ω be a subprobability measure on $A \times X$ satisfying $\omega(\{a\} \times X) \leq \mu(a)q(a)$ for every $a \in A$. Consider the loop that is entered at (a, x) with probability $\omega(a, x)$ and samples $b \leftarrow \nu$ until $I(a, b) = 1$. Then the entry-weighted expected number of samples is at most 1, and for every $\text{Bad} \subseteq X \times B$ the probability that the loop is entered and the selected b satisfies $(x, b) \in \text{Bad}$ is at most*

$$\kappa := \sup_{a \in A, x \in X} \Pr_{b \leftarrow \nu} [I(a, b) = 1 \wedge (x, b) \in \text{Bad}].$$

Proof. An entry with $q(a) = 0$ has $\omega(\{a\} \times X) = 0$, so only entries with $q(a) > 0$ occur. At such an entry the loop makes $1/q(a)$ expected samples and selects b with probability $\nu(b)I(a, b)/q(a)$. The entry-weighted expected number of samples is

$$\sum_{a, x} \frac{\omega(a, x)}{q(a)} = \sum_a \frac{\omega(\{a\} \times X)}{q(a)} \leq \sum_a \mu(a) = 1,$$

and the entered-and-bad mass is

$$\sum_{a, x} \omega(a, x) \frac{\Pr_{b \leftarrow \nu} [I(a, b) = 1 \wedge (x, b) \in \text{Bad}]}{q(a)} \leq \kappa \sum_{a, x} \frac{\omega(a, x)}{q(a)} \leq \kappa. \quad \square$$

Lemma 22 (Resampling One Coordinate, Adapted from [WF26, Lemmas 3, 9 and 10]). *Let D be a distribution on a finite set S , let ν be a distribution on a finite set G , let $k \in \mathbb{N}$, $i \in [k]$, and let $I: G \times S^k \rightarrow \{0, 1\}$. For $\mathbf{s} \in S^k$ and $u \in S$ write $\mathbf{s}^{(i \leftarrow u)}$ for $\mathbf{s}$ with coordinate i replaced by u , and put $p_i(g, \mathbf{s}) := \Pr_{u \leftarrow S} [I(g, \mathbf{s}^{(i \leftarrow u)}) = 1]$, which depends on $\mathbf{s}$ only through the remaining coordinates. Consider the loop that is entered at $(g, \mathbf{s})$ with probability at most $\nu(g)D^k(\mathbf{s})I(g, \mathbf{s})$ and samples $u \leftarrow S$ until $I(g, \mathbf{s}^{(i \leftarrow u)}) = 1$. Then the entry-weighted expected number of samples is at most 1, and for every family of sets $\text{Bad}_{g, \mathbf{s}} \subseteq S \times S$ depending only on $(g, \mathbf{s}_{-i})$ the probability that the loop is entered and the selected u satisfies $(s_i, u) \in \text{Bad}_{g, \mathbf{s}}$ is at most*

$$\kappa := \sup_{g, \mathbf{s}, a \in S} \Pr_{u \leftarrow S} [I(g, \mathbf{s}^{(i \leftarrow u)}) = 1 \wedge (a, u) \in \text{Bad}_{g, \mathbf{s}}].$$

Proof. An entry at $(g, \mathbf{s})$ requires $I(g, \mathbf{s}) = 1$ and, because the entry weight is dominated by $\nu(g)D^k(\mathbf{s})I(g, \mathbf{s})$, also $D(s_i) > 0$. Hence $p_i(g, \mathbf{s}) \geq D(s_i)I(g, \mathbf{s}^{(i \leftarrow s_i)}) = D(s_i) > 0$, so every entered loop terminates. Writing the entry point as $(g, \mathbf{s}_{-i}, a)$ with $a = s_i$, its weight is at most $\nu(g)D^{k-1}(\mathbf{s}_{-i})D(a)I(g, \mathbf{s})$, and the loop makes $1/p_i(g, \mathbf{s})$ expected samples. Since

$$\sum_{a \in S} D(a)I(g, \mathbf{s}^{(i \leftarrow a)}) = p_i(g, \mathbf{s}),$$

summing the entry weights over all values of coordinate i cancels the retry cost. Interpreting terms with a vanishing numerator as zero,

$$\sum_{g, \mathbf{s}_{-i}} \nu(g) D^{k-1}(\mathbf{s}_{-i}) \sum_{a \in S} \frac{D(a) I(g, \mathbf{s}^{(i \leftarrow a)})}{p_i(g, \mathbf{s})} \leq \sum_{g, \mathbf{s}_{-i}} \nu(g) D^{k-1}(\mathbf{s}_{-i}) = 1.$$

For the bad mass, the selected u has law $D(u) I(g, \mathbf{s}^{(i \leftarrow u)}) / p_i(g, \mathbf{s})$, and the inner sum over u is at most κ by definition of the supremum, so the entered-and-bad mass is at most

$$\sum_{g, \mathbf{s}_{-i}} \nu(g) D^{k-1}(\mathbf{s}_{-i}) \sum_{a \in S} D(a) I(g, \mathbf{s}^{(i \leftarrow a)}) \frac{\kappa}{p_i(g, \mathbf{s})} \leq \kappa \sum_{g, \mathbf{s}_{-i}} \nu(g) D^{k-1}(\mathbf{s}_{-i}) = \kappa. \quad \square$$

The hypothesis $\omega(\{a\} \times X) \leq \mu(a)q(a)$ of Lemma 21 is what admits a stored challenge. The probability that a challenge a is stored is the probability that the first run draws a and succeeds at it, which is exactly $\mu(a)q(a)$, whatever the conditional law of the stored challenge looks like. The record x carries whatever the bad events depend on, for instance a candidate tuple extracted before the loop is entered. Lemma 22 is the same statement for a loop that changes one coordinate. There a pointwise bound is impossible, since a single stored challenge may have an arbitrarily small probability of success under a change of one coordinate, and only the sum over all values of that coordinate cancels. The same hypothesis also covers a loop that is started inside a selected execution. Its entry probability is the probability of storing the challenge times the probability of selecting that execution, and the product is at most the probability that the execution is drawn and succeeds, as required by Lemma 22. In Lemma 22 the supremum κ will be taken over the event that the new coordinate does not differ from the old one by a unit. Definition 11 bounds exactly this quantity, with the old value fixed and the new one uniform: for every a , $\Pr_{u \leftarrow \mathcal{C}}[u - a \notin \mathcal{R}_q^\times] \leq \epsilon_C + 1/|\mathcal{C}|$, where $1/|\mathcal{C}|$ accounts for $u = a$.

A.4 The new binding step

Proposition 3 (Binding at a Common Challenge, Adapted from [WF26, Theorem 4]). *Assume the hypotheses of Theorem 1. Then Theorem 1 holds with*

$$\beta_{\text{sis}} = 6\Gamma_C \beta'_{\text{out}}$$

and with the knowledge error obtained from κ by replacing $k/|\mathcal{C}| + k\epsilon_C$ with $2k(1/|\mathcal{C}| + \epsilon_C)$.

Proof. As in the proof of Theorem 1, fix the coins of the prover and let p be the probability that its output instance and witness lie in $\Xi_{\mathcal{R}, q, m, n, \beta'_{\text{out}}}^{\text{lin}}$. Call an execution *successful* if this event holds. Success therefore entails that the verifier completed the norm, trace, and terminal sumcheck checks, which the sumcheck-and-projection stage below uses. Decompose the verifier coins of one execution as $(C, \mathbf{z})$, where C collects the projection, batching, and sumcheck coins and $\mathbf{z} \leftarrow \mathcal{C}^k$ is the folding challenge, and write $I(C, \mathbf{z}) \in \{0, 1\}$ for the success indicator.

The extractor runs one execution on uniform coins. If it is not successful, the extractor stops. Otherwise it stores $(C, \mathbf{z}^{(k)})$ and, for every $i \in [k]$, resamples coordinate i of $\mathbf{z}^{(k)}$ until success, keeping C and the other coordinates fixed. This gives the first family. It then samples fresh coins C' until the execution at $(C', \mathbf{z}^{(k)})$ is successful, and inside this execution builds the second family around the same $\mathbf{z}^{(k)}$ in the same way. Lemma 21 applies to the loop over C' with $A = \mathcal{C}^k$, μ the uniform law of $\mathbf{z}$, $b = C'$, and entry weight $\omega(\mathbf{z}, x) = \Pr[\text{the first execution succeeds at } \mathbf{z} \text{ with record } x]$, where the record x is the transcript of the first execution together with its k coordinate loops, whose marginal on A is exactly $\mu(\mathbf{z}) \Pr_C[I(C, \mathbf{z}) = 1]$. That loop therefore costs at most one expected invocation. Lemma 22 applies to each of the $2k$ coordinate loops, with D the uniform

law on $\mathcal{C}$, $g = C'$ for the first family and $g = C''$ for the second, so each costs at most one expected invocation. For the second family the entry weight at $(C'', \mathbf{z}^{(k)})$ is the probability of storing $\mathbf{z}^{(k)}$ times the probability $\Pr[C''] I(C'', \mathbf{z}^{(k)}) / \Pr_{C''}[I(C'', \mathbf{z}^{(k)}) = 1]$ that the loop selects C'' , which is at most $\Pr[C'] D^k(\mathbf{z}^{(k)}) I(C', \mathbf{z}^{(k)})$, as Lemma 22 requires. In total the extractor makes at most $2k + 2$ expected invocations.

A coordinate loop whose selected value equals the stored one or differs from it by a non-unit is a bad event of Lemma 22. Its supremum is at most $1/|\mathcal{C}| + \epsilon_C$ by Definition 11, so the two families lose $2k(1/|\mathcal{C}| + \epsilon_C)$ in total. Outside this event all $2k$ challenge differences are units. Every execution used below is successful, so its output witness has norm at most β'_{out} and satisfies the folded output equations at its own challenge. The two families thus supply the hypotheses of Lemma 20 at the common challenge $\mathbf{z}^{(k)}$, with the fork witnesses $\mathbf{u}^{(i)}$ and fork values $z_i^{(i)}$ in the roles of $\mathbf{v}_i$ and ζ_i .

In cases (i) and (ii) of Lemma 20 the extractor outputs the SIS-break witness, of norm at most $2\beta'_{\text{out}}$ or $6\Gamma_C\beta'_{\text{out}}$. The latter dominates since $\Gamma_C \geq 1$. In case (iii) the two families define one common tuple $(\mathbf{u}^{(k)}, (\mathbf{w}_i^*)_{i \in [k]})$ and the common accumulator opening $\mathbf{w}_{\text{acc}}^*$. The folded output equations of the second execution give $\text{MLE}[\mathbf{u}^{(k)}](\hat{\mathbf{r}}_{d,1}) = \hat{t}_{\text{acc}} + \sum_{i \in [k]} z_i^{(k)} \hat{t}_i \bmod q$, and the output relations of its forks give $\text{MLE}[\mathbf{w}_i^*](\hat{\mathbf{r}}_{d,1}) = \hat{t}_i \bmod q$, so $\text{MLE}[\mathbf{w}_{\text{acc}}^*](\hat{\mathbf{r}}_{d,1}) = \hat{t}_{\text{acc}} \bmod q$. The candidate tuple is determined by the record of the first execution and hence fixed before the coins C' are drawn. The equalities just derived identify its terminal evaluation with the evaluation the second execution accepted. Each bad event of the sumcheck-and-projection stage is therefore a function of the record and of C' alone, and that stage together with the additive accumulator reconstruction of the proof of Theorem 1 applies verbatim to the second execution. Each of its bad events is charged through Lemma 21, with the record carrying the candidate tuple, and the supremum there is the pointwise bound of the proof of Theorem 1 for uniform coins. This contributes $\kappa_{\text{proj}} + (\log_2 N_d/q)^\mu + 1/q^e + (2 \sum_{j \in [d]} \nu_j + 1)/q^a$ and returns the fresh witnesses of norm at most β'_{in} together with the accumulator witness of norm at most β'_{acc} .

The union bound over the listed bad events gives the stated knowledge error. $\square$